

Fluid Mechanics

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All errors, typographical and substantive, and other offenses, are entirely my own.

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2 - Cartesian Tensors

Comment on notation. To denote the i -th component of a vector, say $\nabla \times \mathbf{v}$, I interchange with the author's notation

$$[\nabla \times \mathbf{v}]_i \iff (\nabla \times \mathbf{v}) \cdot \mathbf{e}_i$$

Question: 2.1

For three spatial dimensions, rewrite the following expressions in index notation and evaluate or simplify them using the values or parameters given, and the definitions of δ_{ij} and ϵ_{ijk} wherever possible. In b) through e), $\mathbf{x}$ is the position vector, with components x_i .

- a. $\mathbf{b} \cdot \mathbf{c}$ where $\mathbf{b} = (1, 4, 17)$ and $\mathbf{c} = (-4, -3, 1)$.
- b. $(\mathbf{u} \cdot \nabla) \mathbf{x}$ where $\mathbf{u}$ a vector with components u_i .
- c. $\nabla \phi$, where $\phi = \mathbf{h} \cdot \mathbf{x}$ and $\mathbf{h}$ is a constant vector with components h_i .
- d. $\nabla \times \mathbf{u}$, where $\mathbf{u} = \boldsymbol{\Omega} \times \mathbf{x}$ and $\boldsymbol{\Omega}$ is a constant vector with components Ω_i .
- e. $\mathbf{C} \cdot \mathbf{x}$, where

$$\mathbf{C} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

a.

$$\begin{aligned} \mathbf{b} \cdot \mathbf{c} &= b_1 c_1 + b_2 c_2 + b_3 c_3 = \sum_i b_i c_i \equiv b_i c_i \\ &= -4 - 12 + 17 \\ &= 1 \end{aligned}$$

b.

$$\begin{aligned} (\mathbf{u} \cdot \nabla) \mathbf{x} &= \left(u_1 \frac{\partial}{\partial x_1} + u_2 \frac{\partial}{\partial x_2} + u_3 \frac{\partial}{\partial x_3} \right) (x_1, x_2, x_3) \\ &= (u_1, u_2, u_3) \\ [(\mathbf{u} \cdot \nabla) \mathbf{x}]_j &= u_i \frac{\partial x_j}{\partial x_i} = \delta_{ij} u_i = u_j \end{aligned}$$

c. Compute $\mathbf{h} \cdot \mathbf{x} = h_1 x_1 + h_2 x_2 + h_3 x_3$.

$$\begin{aligned} \nabla \phi &= \nabla(\mathbf{h} \cdot \mathbf{x}) \\ &= \left(\frac{\partial}{\partial x_1}(\mathbf{h} \cdot \mathbf{x}), \frac{\partial}{\partial x_2}(\mathbf{h} \cdot \mathbf{x}), \frac{\partial}{\partial x_3}(\mathbf{h} \cdot \mathbf{x}) \right) = (h_1, h_2, h_3) \\ [\nabla \phi]_j &= \frac{\partial}{\partial x_j} \sum_i h_i x_i = \delta_{ij} h_i = h_j \end{aligned}$$

d. First compute

$$[\mathbf{u}]_i = [\boldsymbol{\Omega} \times \mathbf{x}]_i = \sum_j \sum_k \epsilon_{ijk} \Omega_j x_k \equiv \epsilon_{ijk} \Omega_j x_k$$

Then we can derive

$$\begin{aligned} [\nabla \times \mathbf{u}]_l &= \sum_m \sum_n \epsilon_{lmn} \frac{\partial u_n}{\partial x_m} \equiv \epsilon_{lmn} \frac{\partial u_n}{\partial x_m} \\ &= \epsilon_{lmn} \frac{\partial}{\partial x_m} (\epsilon_{njk} \Omega_j x_k) \end{aligned}$$

Note: it is nonstandard to have **more than two** repeating indices. Free indices should only appear **once**. Remember these as sanity checks!

e.

$$\begin{aligned} x'_i &= C_{i1}x_1 + C_{i2}x_2 + C_{i3}x_3 = \sum_j C_{ij}x_j \equiv C_{ij}x_j \\ \mathbf{C} \cdot \mathbf{x} &= \begin{bmatrix} x_1 + 2x_2 + 3x_3 \\ x_2 + 2x_3 \\ x_3 \end{bmatrix} \end{aligned}$$

Question: 2.6

Convert $\nabla \times \nabla \rho$ to indicial notation and show that it is zero in Cartesian coordinates for any twice-differentiable scalar function ρ .

We have from equations (2.22) and (2.24)

$$[\nabla \times \nabla \rho]_i = \sum_j \sum_k \epsilon_{ijk} \frac{\partial}{\partial x_j} \left(\frac{\partial \rho}{\partial x_k} \right) \equiv \epsilon_{ijk} \frac{\partial^2 \rho}{\partial x_j \partial x_k}$$

Now, for (i, j, k) permutations of $(1, 2, 3)$, fixed i , and dummy indices l, m , we have surviving terms

$$\epsilon_{ilm} \left(\frac{\partial^2 \rho}{\partial x_l \partial x_m} - \epsilon_{iml} \frac{\partial^2 \rho}{\partial x_m \partial x_l} \right)$$

which we have by Clairaut's theorem, $\partial^2 \rho / \partial x_l \partial x_m = \partial^2 \rho / \partial x_m \partial x_l$ and $\epsilon_{ilm} = -\epsilon_{iml}$. Therefore the surviving terms cancel, leaving $[\nabla \times \nabla \rho]_i = 0$; by arbitrary choice of i we establish $\nabla \times \nabla \rho = \mathbf{0}$.

Question: 2.7

Using indicial notation, show that $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$. [Hint: Call $\mathbf{d} \equiv \mathbf{b} \times \mathbf{c}$. Then $(\mathbf{a} \times \mathbf{d}) \cdot \mathbf{e}_m = \epsilon_{pqm} a_p \epsilon_{ijq} b_i c_j$. Using (2.19), show that $(\mathbf{a} \times \mathbf{d}) \cdot \mathbf{e}_m = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} \cdot \mathbf{e}_m - (\mathbf{a} \cdot \mathbf{b})\mathbf{c} \cdot \mathbf{e}_m$.]

Define $\mathbf{d}$ as above. Introduce dummy indices i and j and free index k . Then the k -th component of $\mathbf{d}$ is

$$d_k \equiv [\mathbf{b} \times \mathbf{c}]_k = \epsilon_{ijk} b_i c_j$$

Next, introduce dummy indices p and q and free index m . The m -th component of $\mathbf{a} \times \mathbf{d} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ is

$$[\mathbf{a} \times \mathbf{d}]_m = \epsilon_{pqm} a_p \epsilon_{ijq} b_i c_j$$

By (2.19), rewrite the product of the epsilons as

$$\epsilon_{ijq} \epsilon_{pqm} = \epsilon_{ijq} \epsilon_{qmp} = \delta_{im} \delta_{jp} - \delta_{ip} \delta_{jm}$$

which leaves us with

$$\begin{aligned} [\mathbf{a} \times (\mathbf{b} \times \mathbf{c})]_m &= (\delta_{im} \delta_{jp} - \delta_{ip} \delta_{jm}) a_p b_i c_j \\ &= (\delta_{jp} a_p c_j) b_i - (\delta_{ip} a_p b_i) c_m \\ &= [(\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}]_m \end{aligned}$$

Question: 2.9

Prove the following relationships: $\delta_{ij} \delta_{ij} = 3$, $\epsilon_{pqr} \epsilon_{pqr} = 6$, and $\epsilon_{pqi} \epsilon_{pqj} = 2\delta_{ij}$.

For fixed indices:

- (a) Note that either δ_{ij} is 0 or 1, so we may reduce $\delta_{ij} \delta_{ij}$ to δ_{ij} .
- (b) Extending this logic a bit, either ϵ_{pqr} is -1 , 0 , or 1 . The product $\epsilon_{pqr} \epsilon_{pqr}$ is either 0 or 1, so we may reduce to $|\epsilon_{pqr}|$.
- (c) For fixed i and ϵ_{pqi} non-zero, the pair (p, q) has $2! = 2$ orderings. Now, suppose $i \neq j$. Keeping the constraint that $\epsilon_{pqj} \neq 0$ also, we find ourselves stuck, as j cannot equal any of p , q , or i . Thus a non-zero term forces $i = j$. From our earlier fact that (p, q) has two permutations, $\epsilon_{pqi} \epsilon_{pqj}$ has two non-zero index configurations for fixed $i = j$.

Using these insights, sum over all index values in each identity to derive

$$\begin{aligned} \delta_{ij} \delta_{ij} &= \delta_{ij} = \delta_{11} + \delta_{22} + \delta_{33} = 3 \\ \epsilon_{pqr} \epsilon_{pqr} &= |\epsilon_{pqr}| = |S_3| = 6 \\ \epsilon_{pqi} \epsilon_{pqj} &= 2\delta_{ij} \end{aligned}$$

where $|S_3| = 3!$ is the cardinality of S_3 , the symmetric group of permutations of $\{1, 2, 3\}$.

Question: 2.10

Show that $\mathbf{C} \cdot \mathbf{C}^T = \mathbf{C}^T \cdot \mathbf{C} = \boldsymbol{\delta}$, where $\mathbf{C}$ is the direction cosine matrix and $\boldsymbol{\delta}$ is the matrix of the Kronecker delta. Any matrix obeying such a relationship is called an *orthogonal matrix* because it represents transformation of one set of orthogonal axes into another.

Derive $[\mathbf{C} \cdot \mathbf{C}^T]_{ij}$ as

$$[\mathbf{C} \cdot \mathbf{C}^T]_{ij} = C_{ik} C_{jk}$$

and $[\mathbf{C}^T \cdot \mathbf{C}]_{ij}$ as

$$[\mathbf{C}^T \cdot \mathbf{C}]_{ij} = C_{ki} C_{kj}$$

Using the definition that $C_{ij} = \mathbf{e}_i \cdot \mathbf{e}'_j$ for unit vectors in the original and rotated coordinate system, where we clarify that the k -th component of $\mathbf{e}_i$ in the rotated coordinate system is $[\mathbf{e}_i]'_k = \mathbf{e}_i \cdot \mathbf{e}'_k$.

$$\begin{aligned}
 C_{ik}C_{jk} &= \sum_k (\mathbf{e}_i \cdot \mathbf{e}'_k)(\mathbf{e}_j \cdot \mathbf{e}'_k) \\
 &= \sum_k [\mathbf{e}_i]'_k [\mathbf{e}_j]'_k \\
 &= \mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij} \\
 C_{ki}C_{kj} &= \sum_k (\mathbf{e}_k \cdot \mathbf{e}'_i)(\mathbf{e}_k \cdot \mathbf{e}'_j) \\
 &= \sum_k [\mathbf{e}'_i]_k [\mathbf{e}'_j]_k \\
 &= \mathbf{e}'_i \cdot \mathbf{e}'_j = \delta_{ij}
 \end{aligned}$$

where we appeal to the invariance of the dot product under coordinate rotation, proving the desired result.

Question: 2.11

Show that for a second-order tensor $\mathbf{A}$, the following quantities are invariant under a rotation of axes:

$$I_1 = A_{ii}$$

$$I_2 = \det \begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix} + \det \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} + \det \begin{vmatrix} A_{11} & A_{13} \\ A_{31} & A_{33} \end{vmatrix}, \text{ and}$$

$$I_3 = \det(A_{ij})$$

[Hint: Use the result of Exercise 2.10 and the transformation rule (2.12) to show that $I'_1 = A'_{ii} = A_{ii} = I_1$. Then show that $A_{ij}A_{ji}$ and $A_{ij}A_{jk}A_{ki}$ are also invariants. In fact, *all* contracted scalars of the form $A_{ij}A_{jk} \cdots A_{mi}$ are invariants. Finally, verify that $I_2 = \frac{1}{2}[I_1^2 - A_{ij}A_{ji}]$, and $I_3 = \frac{1}{3}[A_{ij}A_{jk}A_{ki} - I_1A_{ij}A_{ji} + I_2A_{ii}]$. Because the right-hand sides are invariant, so are I_2 and I_3 .]

By the transformation rule (2.12) and exercise 2.10,

$$I'_1 = A'_{ii} = \underbrace{C_{mi}C_{ni}}_{\delta_{mn}} A_{mn} = \delta_{mn} A_{mn}$$

Now, $\delta_{mn} = 1$ only when $m = n$, so we may set both equal to i and conclude

$$I'_1 = A'_{ii} = A_{ii} = I_1$$

We call I_1 the **trace**. In general, we can show that $A_{ij}A_{ji}$ is also invariant. Similarly, write from the transformation rule

$$\begin{aligned}
 A'_{ij}A'_{ji} &= C_{mi}C_{nj}A_{mn}C_{kj}C_{li}A_{kl} \\
 &= \underbrace{C_{mi}C_{li}}_{\delta_{ml}} A_{mn} \underbrace{C_{nj}C_{kj}}_{\delta_{nk}} A_{kl}
 \end{aligned}$$

where a non-zero product is guaranteed only when $m = l$ and $n = k$. Set $m = l = i$ and $n = k = j$ to conclude $A'_{ij}A'_{ji} = A_{ij}A_{ji}$.

For a finite number of terms, we can prove $A'_{ij}A'_{jk} \cdots A'_{mi} = A_{ij}A_{jk} \cdots A_{mi}$. Do so by observing

$$A'_{ij}A'_{jk} \cdots A'_{mi} = \textcolor{red}{C}_{oi}\textcolor{blue}{C}_{nj}A_{on}\textcolor{blue}{C}_{fj}\textcolor{green}{C}_{gk}A_{fg} \cdots C_{pm}\textcolor{red}{C}_{qi}A_{pq}$$

By pairing terms of the same color, we can collapse the product into sequences of $C_{ik}C_{jk} = \delta_{ij}$, which when applied with the aforementioned logic, produce the "chain" of A 's where the second index of one term is the first index of the successive term.

For the final phase, we show that

$$I_2 = \frac{1}{2}[I_1^2 - A_{ij}A_{ji}]$$

From the definition of I_2 , compute

$$\begin{aligned} I_2 &= (A_{11}A_{22} - A_{12}A_{21}) + (A_{22}A_{33} - A_{23}A_{32}) + (A_{11}A_{33} - A_{13}A_{31}) \\ &= \frac{1}{2} \sum_i \sum_j (A_{ii}A_{jj} - A_{ij}A_{ji}) \\ &= \frac{1}{2} (A_{ii}A_{jj} - A_{ij}A_{ji}) \\ &= \frac{1}{2} (I_1^2 - A_{ij}A_{ji}) \end{aligned}$$

where the $1/2$ term arises from double counting in the double sum whenever $i \neq j$. In the case when $i = j$, the term vanishes. From the fact that each term on the right-hand side of I_2 is invariant under transformation, we conclude $I'_2 = I_2$.

Lastly, for I_3 , note that the determinant of a 3-by-3 matrix is given by

$$\frac{1}{6} \epsilon_{ijk} \epsilon_{pqr} A_{ip} A_{jq} A_{kr}$$

To simplify the product of permutation symbols, we introduce an alternate definition, expressed in terms of the determinant of a matrix of Kronecker deltas:

$$\epsilon_{ijk} = \det \begin{vmatrix} \mathbf{e}_i \\ \mathbf{e}_j \\ \mathbf{e}_k \end{vmatrix} = \det \begin{vmatrix} \delta_{i1} & \delta_{i2} & \delta_{i3} \\ \delta_{j1} & \delta_{j2} & \delta_{j3} \\ \delta_{k1} & \delta_{k2} & \delta_{k3} \end{vmatrix}$$

The product is thus the determinant of the matrix product

$$\begin{aligned} \epsilon_{ijk} \epsilon_{pqr} &= \det \begin{vmatrix} \mathbf{e}_i \cdot \mathbf{e}_p & \mathbf{e}_i \cdot \mathbf{e}_q & \mathbf{e}_i \cdot \mathbf{e}_r \\ \mathbf{e}_j \cdot \mathbf{e}_p & \mathbf{e}_j \cdot \mathbf{e}_q & \mathbf{e}_j \cdot \mathbf{e}_r \\ \mathbf{e}_k \cdot \mathbf{e}_p & \mathbf{e}_k \cdot \mathbf{e}_q & \mathbf{e}_k \cdot \mathbf{e}_r \end{vmatrix} \\ &= \det \begin{vmatrix} \delta_{ip} & \delta_{iq} & \delta_{ir} \\ \delta_{jp} & \delta_{jq} & \delta_{jr} \\ \delta_{kp} & \delta_{kq} & \delta_{kr} \end{vmatrix} \\ &= \delta_{ip}(\delta_{jq}\delta_{kr} - \delta_{jr}\delta_{kq}) - \delta_{iq}(\delta_{jp}\delta_{kr} - \delta_{jr}\delta_{kp}) + \delta_{ir}(\delta_{jp}\delta_{kq} - \delta_{jq}\delta_{kp}) \end{aligned}$$

The next step is some accounting tedium. Essentially, we must figure out all of the ways we can set the triplet (p, q, r) equal to (i, j, k) . There are six such ways

of doing so: in the current order, set equal by holding position, and then “shifting” once to the left or once to the right. The other three arise from flipping the order of the first triplet, and once more holding or shifting; these beget the sign change as well from flipping the order of the Levi-Civita indices.

If you are familiar with the language of dihedral groups, particularly the isomorphism $S_3 \cong D_3$ (relating the structure of the symmetric group of three elements and rotation and reflection of a triangle), you will notice some similarities here. Our next task is to evaluate the Levi-Civita product term by term and reduce.

Term	Simplification
$\frac{1}{6} \delta_{ip} \delta_{jq} \delta_{kr} A_{ip} A_{jq} A_{kr}$	$\frac{1}{6} A_{ii} A_{jj} A_{kk}$
$-\frac{1}{6} \delta_{ip} \delta_{jr} \delta_{kq} A_{ip} A_{jq} A_{kr}$	$-\frac{1}{6} A_{ii} A_{jk} A_{kj}$
$-\frac{1}{6} \delta_{iq} \delta_{jp} \delta_{kr} A_{ip} A_{jq} A_{kr}$	$-\frac{1}{6} A_{ij} A_{ji} A_{kk}$
$\frac{1}{6} \delta_{iq} \delta_{jr} \delta_{kp} A_{ip} A_{jq} A_{kr}$	$\frac{1}{6} A_{ik} A_{ji} A_{kj}$
$\frac{1}{6} \delta_{ir} \delta_{jp} \delta_{kq} A_{ip} A_{jq} A_{kr}$	$\frac{1}{6} A_{ij} A_{jk} A_{ki}$
$-\frac{1}{6} \delta_{ir} \delta_{jq} \delta_{kp} A_{ip} A_{jq} A_{kr}$	$-\frac{1}{6} A_{ik} A_{jj} A_{ki}$

Adding the six terms, which we color-code by virtue of structure of the indices, produces (which we set equal to I_3 as at the outset we began with the indicial formula for $\det(A_{ij})$)

$$I_3 = \frac{1}{6} (I_1^3 - 3A_{ii} A_{jk} A_{kj} + 2A_{ij} A_{jk} A_{ki})$$

The first term of the text's form of I_3 is the $(1/3) A_{ij} A_{jk} A_{ki}$ term here. The remaining two terms can be split up to reveal

$$\begin{aligned}
 A_{ii} A_{jj} A_{kk} - 3A_{ii} A_{jk} A_{kj} &= \underbrace{A_{ii} A_{jj} A_{kk} - A_{ii} A_{jk} A_{kj}}_{2A_{ii}(A_{jj} A_{kk} - A_{jk} A_{kj})} - \underbrace{2A_{ii} A_{jk} A_{kj}}_{2I_1 A_{ij} A_{ji}} \\
 &= 2(I_2 A_{ii} - I_1 A_{ij} A_{ji})
 \end{aligned}$$

which takes us to the text's form of I_3 . At long last, we have proven that every term of I_3 is invariant under transformation, and thus $I'_3 = I_3$.

Question: 2.13

Show that δ_{ij} is an isotropic tensor. That is, show that $\delta'_{ij} = \delta_{ij}$ under rotation of the coordinate system. [Hint: Use the transformation rule (2.12) and the results of Exercise 2.10.]

By the transformation rule (2.12),

$$\delta'_{ij} = C_{mi}C_{nj}\delta_{mn}$$

which produces non-zero terms only when $m = n$. Set both equal to k and rewrite as

$$\delta'_{ij} = C_{ki}C_{kj}$$

By the result of 2.10, $C_{ki}C_{kj} = \delta_{ij}$, establishing the proof that δ_{ij} is an isotropic tensor.

Question: 2.17

If a is a positive constant and $\mathbf{b}$ is a constant vector, determine the divergence and the curl of $\mathbf{u} = a\mathbf{x}/x^3$ and $\mathbf{u} = \mathbf{b} \times (\mathbf{x}/x^2)$ where $x = \sqrt{x_1^2 + x_2^2 + x_3^2} \equiv \sqrt{x_i x_i}$ is the length of $\mathbf{x}$.

For the divergence, compute

$$\nabla \cdot \mathbf{u} \equiv \frac{\partial u_i}{\partial x_i}$$

where $u_i = ax_i(x_j x_j)^{-3/2}$ implies

$$\begin{aligned} \frac{\partial u_i}{\partial x_i} &= \frac{\partial}{\partial x_i} \left[\frac{ax_i}{(x_j x_j)^{3/2}} \right] \\ &= \frac{a\delta_{ii}}{(x_j x_j)^{3/2}} - \frac{3ax_i x_i}{(x_j x_j)^{5/2}} \\ &= \frac{3a}{(x_j x_j)^{3/2}} - \frac{3a}{(x_j x_j)^{3/2}} \\ &= 0 \quad (x_j \neq 0) \end{aligned}$$

We refer to this function as the **inverse-square radial field**, and it shows up ubiquitously in fluid mechanics and electrodynamics. Its curl is

$$\begin{aligned} [\nabla \times \mathbf{u}]_i &= \epsilon_{ijk} \frac{\partial}{\partial x_j} \left[\frac{ax_k}{(x_l x_l)^{3/2}} \right] \\ &= \epsilon_{ijk} \left[\frac{a}{(x_l x_l)^{3/2}} \frac{\partial x_k}{\partial x_j} - \frac{3ax_k}{2(x_l x_l)^{5/2}} \underbrace{\left(\frac{\partial(x_l x_l)}{\partial x_j} \right)}_{2x_j} \right] \\ &= \epsilon_{ijk} \left[\frac{a\delta_{kj}}{(x_l x_l)^{3/2}} - \frac{3ax_k x_j}{(x_l x_l)^{5/2}} \right] \end{aligned}$$

The first term, which contains $\epsilon_{ijk}\delta_{kj}$, must vanish, for $j = k$ has $\delta_{kj} \neq 0$ but $\epsilon_{ijk} = 0$, and $j \neq k$ keeps $\epsilon_{ijk} \neq 0$ but $\delta_{kj} = 0$. The second term, which has $\epsilon_{ijk}x_kx_j$, vanishes by contraction of the antisymmetry of ϵ_{ijk} and symmetry of x_jx_k . Therefore $[\nabla \times \mathbf{u}]_i = 0$.

The second function is

$$u_i = \epsilon_{ijk} \frac{b_j x_k}{x_l x_l}$$

which has divergence

$$\begin{aligned} \nabla \cdot \mathbf{u} &= \frac{\partial}{\partial x_i} \left(\epsilon_{ijk} \frac{b_j x_k}{x_l x_l} \right) \\ &= \epsilon_{ijk} \left[\frac{b_j}{x_l x_l} \frac{\partial x_k}{\partial x_i} - \frac{b_j x_k}{(x_l x_l)^2} \frac{\partial (x_l x_l)}{\partial x_i} \right] \\ &= \epsilon_{ijk} \left[\frac{b_j \delta_{ki}}{x_l x_l} - \frac{2b_j x_k x_i}{(x_l x_l)^2} \right] \end{aligned}$$

In the first term, $\epsilon_{ijk}\delta_{ki} = 0$ for any configuration of indices, and by the symmetry-antisymmetry argument, $\epsilon_{ijk}x_kx_i$ in the second term implies that the second term vanishes too. Thus $\nabla \cdot \mathbf{u} = 0$. Lastly, the curl

$$\begin{aligned} [\nabla \times \mathbf{u}]_i &= \epsilon_{ijk} \frac{\partial}{\partial x_j} \left(\epsilon_{klm} \frac{b_l x_m}{x_n x_n} \right) \\ &= \epsilon_{ijk} \epsilon_{klm} \left[\frac{b_l}{x_n x_n} \frac{\partial x_m}{\partial x_j} - \frac{b_l x_m}{(x_n x_n)^2} \frac{\partial (x_n x_n)}{\partial x_j} \right] \\ &= \epsilon_{ijk} \epsilon_{klm} \left[\frac{b_l \delta_{mj}}{x_n x_n} - \frac{2b_l x_m x_j}{(x_n x_n)^2} \right] \\ &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \left[\frac{b_l \delta_{mj}}{x_n x_n} - \frac{2b_l x_m x_j}{(x_n x_n)^2} \right] \\ &= \frac{b_l \delta_{il} \delta_{jm} \delta_{mj}}{x_n x_n} - \frac{b_l \delta_{im} \delta_{mj} \delta_{jl}}{x_n x_n} - \frac{2b_l x_m x_j \delta_{il} \delta_{jm}}{(x_n x_n)^2} + \frac{2b_l x_m x_j \delta_{im} \delta_{jl}}{(x_n x_n)^2} \\ &= \frac{3b_i}{x_n x_n} - \frac{b_i}{x_n x_n} - \frac{2b_i}{x_n x_n} + \frac{2b_j x_i x_j}{(x_n x_n)^2} \\ &= \frac{2b_j x_i x_j}{(x_n x_n)^2} \end{aligned}$$

Thus the curl is

$$\nabla \times \mathbf{u} = \frac{2(\mathbf{b} \cdot \mathbf{x})}{x^4} \mathbf{x}, \quad \mathbf{x} \neq \mathbf{0}$$

Question: 2.18

Obtain the recipe for the gradient of a scalar function in cylindrical polar coordinates from the integral definition (2.32).

We will rework (2.32) slightly to take in a scalar function. Let $\mathbf{Q} = \phi \mathbf{c}$, where $\mathbf{c}$ is a constant vector and ϕ a scalar function. Using the fact that $\nabla \cdot \phi \mathbf{c} = \mathbf{c} \cdot \nabla \phi$,

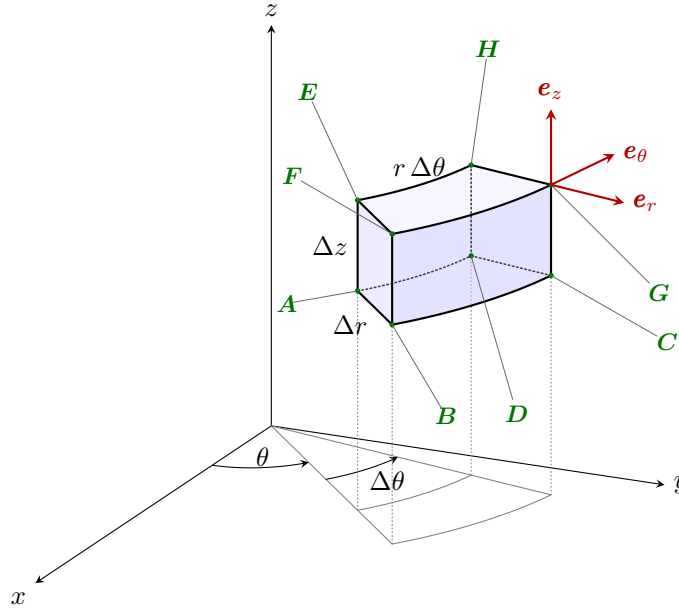
we have

$$\mathbf{c} \cdot \nabla \phi = \lim_{V \rightarrow 0} \mathbf{c} \cdot \left(\frac{1}{V} \iint_A \phi \mathbf{n} dA \right)$$

However, since $\mathbf{c}$ may be any arbitrary constant vector, this grants us license to drop the $\mathbf{c}$ from both sides:

$$\nabla \phi = \lim_{V \rightarrow 0} \frac{1}{V} \iint_A \phi \mathbf{n} dA$$

For our infinitesimal volume, we consider the wedge bounded by $[r, r + \Delta r]$ along $\mathbf{e}_r$, $[\theta, \theta + \Delta \theta]$ along $\mathbf{e}_\theta$, and $[z, z + \Delta z]$ along $\mathbf{e}_z$. Thus the volume is $r \Delta r \Delta \theta \Delta z$. The red unit vectors below denote the positive direction for each surface.



Following the method in example 2.12, begin with Taylor expanding ϕ from the center of the volume to the center of each side; **be aware that this will depend on the specified face**. As an example, we derive for faces $BCFG$ and $ADEH$:

$$\begin{aligned} [\phi]_{BCFG} &= \phi + \frac{\Delta r}{2} \frac{\partial \phi}{\partial r} + \dots \\ [\phi]_{ADEH} &= \phi - \frac{\Delta r}{2} \frac{\partial \phi}{\partial r} + \dots \end{aligned}$$

where the positive and minus signs arise from either a positive or negative displacement from the center of the volume, i.e. in the direction of $\mathbf{n} = \mathbf{e}_r$ or $\mathbf{n} = -\mathbf{e}_r$. Put differently, our use of the midpoint rule on each face enables this clean orientation of the anti-parallel normals along the r -axis. The infinitesimal area on these faces is

$(r - \Delta r/2) \Delta\theta\Delta z$ on $ADEH$ and $(r + \Delta r/2) \Delta\theta\Delta z$ on $BCFG$. Then $\phi \mathbf{n} dA$ is

$$\begin{aligned} ([\phi \mathbf{n}]_{BCFG} + [\phi \mathbf{n}]_{ADEH}) dA &= \left(\mathbf{e}_r \left[\phi + \frac{\Delta r}{2} \frac{\partial \phi}{\partial r} + \dots \right] \left(r + \frac{\Delta r}{2} \right) \right. \\ &\quad \left. - \mathbf{e}_r \left[\phi - \frac{\Delta r}{2} \frac{\partial \phi}{\partial r} + \dots \right] \left(r - \frac{\Delta r}{2} \right) \right) \Delta\theta\Delta z \\ &= \mathbf{e}_r \left(\phi \Delta r + r \Delta r \frac{\partial \phi}{\partial r} + \dots \right) \Delta\theta\Delta z \\ &= \mathbf{e}_r \left(\frac{\phi}{r} + \frac{\partial \phi}{\partial r} + \dots \right) r \Delta r \Delta\theta\Delta z \end{aligned}$$

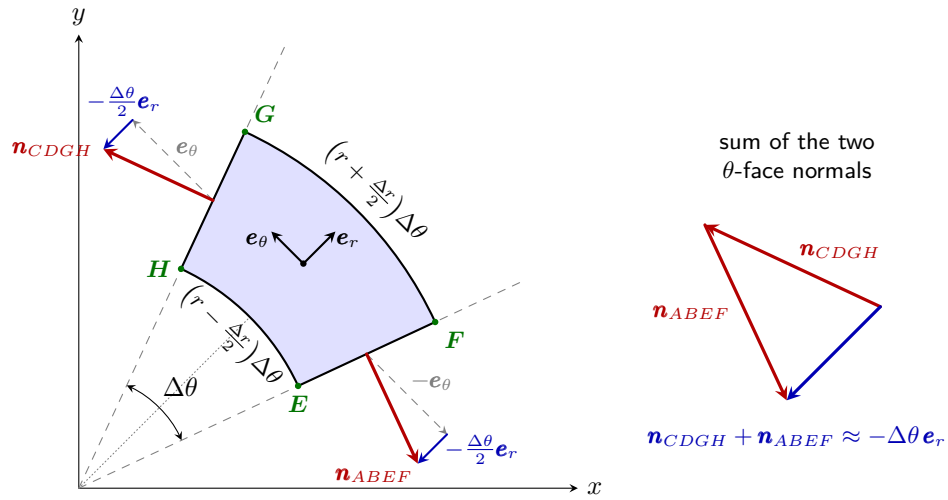
Analogously, we can compute

$$\begin{aligned} ([\phi \mathbf{n}]_{CDGH} + [\phi \mathbf{n}]_{ABEF}) dA &= \left(\left[\mathbf{e}_\theta - \frac{\Delta\theta}{2} \mathbf{e}_r \right] \left[\phi + \frac{\Delta\theta}{2} \frac{\partial \phi}{\partial \theta} + \dots \right] \right. \\ &\quad \left. - \left[\mathbf{e}_\theta + \frac{\Delta\theta}{2} \mathbf{e}_r \right] \left[\phi - \frac{\Delta\theta}{2} \frac{\partial \phi}{\partial \theta} + \dots \right] \right) \Delta r \Delta z \\ &= \left[-\mathbf{e}_r \phi \Delta\theta + \mathbf{e}_\theta \left(\Delta\theta \frac{\partial \phi}{\partial \theta} + \dots \right) \right] \Delta r \Delta z \\ &= \left(-\mathbf{e}_r \frac{\phi}{r} + \mathbf{e}_\theta \frac{1}{r} \frac{\partial \phi}{\partial \theta} \right) r \Delta r \Delta\theta \Delta z \\ ([\phi \mathbf{n}]_{EFGH} + [\phi \mathbf{n}]_{ABCD}) dA &= \left(\mathbf{e}_z \left[\phi + \frac{\Delta z}{2} \frac{\partial \phi}{\partial z} + \dots \right] - \mathbf{e}_z \left[\phi - \frac{\Delta z}{2} \frac{\partial \phi}{\partial z} + \dots \right] \right) r \Delta r \Delta\theta \\ &= \mathbf{e}_z \left(\frac{\partial \phi}{\partial z} + \dots \right) r \Delta r \Delta\theta \Delta z \end{aligned}$$

where the $\mathbf{e}_\theta$ directions warrant some discussion. Observe in the diagram below that the normals of the “opposite” faces along that axis are not necessarily anti-parallel with one another. As it turns out, the normals are pointing away from $\mathbf{e}_\theta$ by some amount; namely, this amount is $\Delta\theta/2$ in the direction of $-\mathbf{e}_r$. From the Taylor series expansion of $\mathbf{e}_\theta$, this looks like

$$\mathbf{e}_\theta \left(\theta \pm \frac{\Delta\theta}{2} \right) = \mathbf{e}_\theta \pm \frac{\Delta\theta}{2} \frac{\partial \mathbf{e}_\theta}{\partial \theta} + \dots = \mathbf{e}_\theta \mp \frac{\Delta\theta}{2} \mathbf{e}_r + \dots$$

where we use the classic result that $\partial \mathbf{e}_\theta / \partial \theta = -\mathbf{e}_r$.



Adding our three results together, dividing by the volume and taking the limit, conclude

$$\begin{aligned}
 \nabla \phi &= \lim_{V \rightarrow 0} \frac{1}{V} \iint_A \phi \mathbf{n} dA \\
 &= \lim_{\substack{\Delta r \rightarrow 0 \\ \Delta \theta \rightarrow 0 \\ \Delta z \rightarrow 0}} \frac{1}{r \Delta r \Delta \theta \Delta z} \left[\mathbf{e}_r \left(\phi + \frac{\partial \phi}{\partial r} \Delta r \right) - \mathbf{e}_r \phi + \mathbf{e}_\theta \frac{1}{r} \frac{\partial \phi}{\partial \theta} \Delta \theta + \mathbf{e}_z \frac{\partial \phi}{\partial z} \Delta z + \dots \right] r \Delta r \Delta \theta \Delta z \\
 &= \mathbf{e}_r \frac{\partial \phi}{\partial r} + \mathbf{e}_\theta \frac{1}{r} \frac{\partial \phi}{\partial \theta} + \mathbf{e}_z \frac{\partial \phi}{\partial z}
 \end{aligned}$$

where we keep only first-order terms in the last equality. Observe there that when ϕ is constant, $\nabla \phi = \mathbf{0}$, comporting with the result $\oint \mathbf{n} dA = \mathbf{0}$.

Question: 2.19

Obtain the recipe for the divergence of a vector function in cylindrical polar coordinates from the integral definition (2.32).

The integral definition of divergence is

$$\nabla \cdot \mathbf{Q} = \lim_{V \rightarrow 0} \frac{1}{V} \iint_A \mathbf{n} \cdot \mathbf{Q} dA$$

Building off of our work in 2.18, derive

$$\begin{aligned}
([\mathbf{n} \cdot \mathbf{Q}]_{BCFG} + [\mathbf{n} \cdot \mathbf{Q}]_{ADEH}) dA &= \left(\mathbf{e}_r \cdot \left[\mathbf{Q} + \frac{\Delta r}{2} \frac{\partial \mathbf{Q}}{\partial r} + \dots \right] \left(r + \frac{\Delta r}{2} \right) \right. \\
&\quad \left. - \mathbf{e}_r \cdot \left[\mathbf{Q} - \frac{\Delta r}{2} \frac{\partial \mathbf{Q}}{\partial r} + \dots \right] \left(r - \frac{\Delta r}{2} \right) \right) \Delta \theta \Delta z \\
&= \mathbf{e}_r \cdot \left(\mathbf{Q} \Delta r + r \Delta r \frac{\partial \mathbf{Q}}{\partial r} + \dots \right) \Delta \theta \Delta z \\
&= \mathbf{e}_r \cdot \left(\frac{\mathbf{Q}}{r} + \frac{\partial \mathbf{Q}}{\partial r} + \dots \right) r \Delta r \Delta \theta \Delta z \\
&= \left(\frac{Q_r}{r} + \frac{\partial Q_r}{\partial r} + \dots \right) r \Delta r \Delta \theta \Delta z \\
([\mathbf{n} \cdot \mathbf{Q}]_{CDGH} + [\mathbf{n} \cdot \mathbf{Q}]_{ABEF}) dA &= \left(\left[\mathbf{e}_\theta - \frac{\Delta \theta}{2} \mathbf{e}_r \right] \cdot \left[\mathbf{Q} + \frac{\Delta \theta}{2} \frac{\partial \mathbf{Q}}{\partial \theta} + \dots \right] \right. \\
&\quad \left. - \left[\mathbf{e}_\theta + \frac{\Delta \theta}{2} \mathbf{e}_r \right] \cdot \left[\mathbf{Q} - \frac{\Delta \theta}{2} \frac{\partial \mathbf{Q}}{\partial \theta} + \dots \right] \right) \Delta r \Delta z \\
&= \left[-\mathbf{e}_r \cdot \mathbf{Q} \Delta \theta + \mathbf{e}_\theta \cdot \left(\Delta \theta \frac{\partial \mathbf{Q}}{\partial \theta} + \dots \right) \right] \Delta r \Delta z \\
&= \left(-\frac{Q_r}{r} + \frac{Q_r}{r} + \frac{1}{r} \frac{\partial Q_\theta}{\partial \theta} + \dots \right) r \Delta r \Delta \theta \Delta z \\
&= \left(\frac{1}{r} \frac{\partial Q_\theta}{\partial \theta} + \dots \right) r \Delta r \Delta \theta \Delta z \\
([\mathbf{n} \cdot \mathbf{Q}]_{EFGH} + [\mathbf{n} \cdot \mathbf{Q}]_{ABCD}) dA &= \left(\mathbf{e}_z \cdot \left[\mathbf{Q} + \frac{\Delta z}{2} \frac{\partial \mathbf{Q}}{\partial z} + \dots \right] - \mathbf{e}_z \cdot \left[\mathbf{Q} - \frac{\Delta z}{2} \frac{\partial \mathbf{Q}}{\partial z} + \dots \right] \right) r \Delta r \Delta \theta \\
&= \left(\frac{\partial Q_z}{\partial z} + \dots \right) r \Delta r \Delta \theta \Delta z
\end{aligned}$$

where a note is needed in the derivation of the integrand along the $\mathbf{e}_\theta$ faces. In particular, it is generally incorrect that

$$\mathbf{e}_\theta \cdot \frac{\partial \mathbf{Q}}{\partial \theta} = \frac{\partial Q_\theta}{\partial \theta}$$

However, referring to the component definition of $\mathbf{Q}$

$$\mathbf{Q} = Q_r \mathbf{e}_r + Q_\theta \mathbf{e}_\theta + Q_z \mathbf{e}_z$$

and differentiating

$$\frac{\partial \mathbf{Q}}{\partial \theta} = Q_r \frac{\partial \mathbf{e}_r}{\partial \theta} + \mathbf{e}_\theta \frac{\partial Q_\theta}{\partial \theta} + Q_\theta \frac{\partial \mathbf{e}_\theta}{\partial \theta} = \left(Q_r + \frac{\partial Q_\theta}{\partial \theta} \right) \mathbf{e}_\theta + \left(\frac{\partial Q_r}{\partial \theta} - Q_\theta \right) \mathbf{e}_r + \frac{\partial Q_z}{\partial \theta} \mathbf{e}_z$$

We can clearly see that

$$\mathbf{e}_\theta \cdot \frac{\partial \mathbf{Q}}{\partial \theta} = Q_r + \frac{\partial Q_\theta}{\partial \theta}$$

This arises from the fact that $\partial \mathbf{e}_r / \partial \theta = \mathbf{e}_\theta$ (namely, that $\mathbf{e}_r$ is defined in terms of θ).

With that out of the way, sum each piece as

$$\begin{aligned}
 \nabla \cdot \mathbf{Q} &= \lim_{V \rightarrow 0} \frac{1}{V} \iint_A \mathbf{n} \cdot \mathbf{Q} dA \\
 &= \lim_{\substack{\Delta r \rightarrow 0 \\ \Delta \theta \rightarrow 0 \\ \Delta z \rightarrow 0}} \frac{1}{r \Delta r \Delta \theta \Delta z} \left[\frac{Q_r}{r} + \frac{\partial Q_r}{\partial r} + \frac{1}{r} \frac{\partial Q_\theta}{\partial \theta} + \frac{\partial Q_z}{\partial z} + \dots \right] r \Delta r \Delta \theta \Delta z \\
 &= \frac{1}{r} \left[\frac{\partial(r Q_r)}{\partial r} + \frac{\partial Q_\theta}{\partial \theta} \right] + \frac{\partial Q_z}{\partial z}
 \end{aligned}$$

Notably, the derivation of the divergence using the integral definition exposes the unequal faces along the r -axis as the source of the Q_r/r that emerges in the divergence.

Question: 2.20

Obtain the recipe for the divergence of a vector function in spherical polar coordinates from the integral definition (2.32).

We illustrate our infinitesimal wedge in spherical coordinates below. The first key conceptual difference to understand is as follows: where the infinitesimal wedge in cylindrical coordinates did not have anti-parallel norms on opposite faces along the θ axis, in spherical coordinates this manifests in two places, the θ and ϕ axes. The infinitesimal volume is thus $r^2 \sin(\theta) \Delta r \Delta \theta \Delta \phi$.

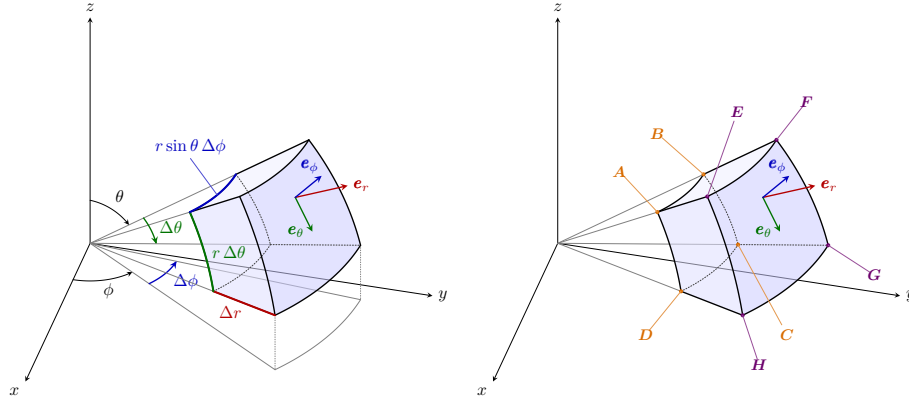


Figure 1: Infinitesimal volume element in spherical coordinates: edge lengths (left) and vertex labels (right).

Once more, apply the midpoint rule as the textbook does, meaning we measure distances relative to the midpoint of the infinitesimal wedge. We begin with the faces along the r -axis, $ABCD$ and $EFGH$. Their areas differ by virtue of being located at different radii along the $\mathbf{e}_r$ unit vector: $r - \Delta r/2$ and $r + \Delta r/2$. The

normals for each face are

$$\begin{aligned}
 [\mathbf{n} dA]_{ABCD} &= -\mathbf{e}_r \left(r - \frac{\Delta r}{2} \right)^2 \sin(\theta) \Delta\theta \Delta\phi \\
 &= \boxed{-\mathbf{e}_r (r^2 - r\Delta r) \sin(\theta) \Delta\theta \Delta\phi} \\
 [\mathbf{n} dA]_{EFGH} &= \mathbf{e}_r \left(r + \frac{\Delta r}{2} \right)^2 \sin(\theta) \Delta\theta \Delta\phi \\
 &= \boxed{\mathbf{e}_r (r^2 + r\Delta r) \sin(\theta) \Delta\theta \Delta\phi}
 \end{aligned}$$

where higher-order terms are dropped. Now we repeat the task along the other two axes, for which our work is a tad more complicated. Begin with the θ faces $ABFE$ and $CDHG$. Recall first the Cartesian coordinate definitions of $\mathbf{e}_r$ and $\mathbf{e}_\theta$:

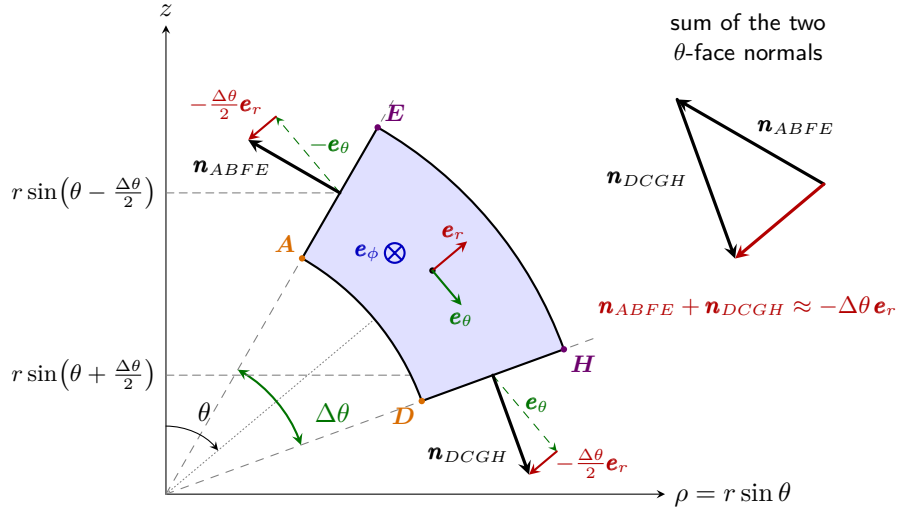
$$\begin{aligned}
 \mathbf{e}_r(\theta, \phi) &= \sin(\theta) \cos(\phi) \hat{\mathbf{x}} + \sin(\theta) \sin(\phi) \hat{\mathbf{y}} + \cos(\theta) \hat{\mathbf{z}} \\
 \mathbf{e}_\theta(\theta, \phi) &= \cos(\theta) \cos(\phi) \hat{\mathbf{x}} + \cos(\theta) \sin(\phi) \hat{\mathbf{y}} - \sin(\theta) \hat{\mathbf{z}}
 \end{aligned}$$

where $\partial \mathbf{e}_\theta / \partial \theta = -\mathbf{e}_r$. Referring to Figure 1 for the position of these faces, we see that $ABFE$ is at $\theta - \Delta\theta/2$ and $CDHG$ at $\theta + \Delta\theta/2$. We may Taylor expand $\mathbf{e}_\theta(\theta \pm \Delta\theta/2, \phi)$ to approximate

$$\mathbf{e}_\theta \left(\theta \pm \frac{\Delta\theta}{2}, \phi \right) = \mathbf{e}_\theta(\theta, \phi) \pm \frac{\Delta\theta}{2} \frac{\partial \mathbf{e}_\theta}{\partial \theta} + \cdots = \mathbf{e}_\theta(\theta, \phi) \mp \frac{\Delta\theta}{2} \mathbf{e}_r + \cdots$$

Ignoring the higher-order terms, look to the schematic below. The normal to face $ABFE$ (seen here from the side along AE) is $-\mathbf{e}_\theta + \mathbf{e}_r(\Delta\theta/2)$, and to $CDHG$, seen from the side along segment DH , is $\mathbf{e}_\theta - \mathbf{e}_r(\Delta\theta/2)$. The respective norms are

$$\begin{aligned}
 [\mathbf{n} dA]_{ABFE} &= -\left(\mathbf{e}_\theta + \mathbf{e}_r \frac{\Delta\theta}{2} \right) r \sin(\theta - \Delta\theta/2) \Delta r \Delta\phi \\
 &= -\left(\mathbf{e}_\theta + \mathbf{e}_r \frac{\Delta\theta}{2} \right) r \left[\sin(\theta) - \frac{\Delta\theta}{2} \cos(\theta) \right] \Delta r \Delta\phi \\
 &= \boxed{\mathbf{e}_\theta \left[-r \sin(\theta) + \frac{r\Delta\theta}{2} \cos(\theta) \right] \Delta r \Delta\phi - \mathbf{e}_r \frac{r\Delta\theta}{2} \sin(\theta) \Delta r \Delta\phi} \\
 [\mathbf{n} dA]_{CDHG} &= \left(\mathbf{e}_\theta - \mathbf{e}_r \frac{\Delta\theta}{2} \right) r \sin(\theta + \Delta\theta/2) \Delta r \Delta\phi \\
 &= \left(\mathbf{e}_\theta - \mathbf{e}_r \frac{\Delta\theta}{2} \right) r \left[\sin(\theta) + \frac{\Delta\theta}{2} \cos(\theta) \right] \Delta r \Delta\phi \\
 &= \boxed{\mathbf{e}_\theta \left[r \sin(\theta) + \frac{r\Delta\theta}{2} \cos(\theta) \right] \Delta r \Delta\phi - \mathbf{e}_r \frac{r\Delta\theta}{2} \sin(\theta) \Delta r \Delta\phi}
 \end{aligned}$$



Lastly, we compute the normals for the ϕ faces. The faces $ADHE$ and $BCGF$ are at along the azimuthal angles $\phi - \Delta\phi/2$ and $\phi + \Delta\phi/2$, respectively. Analogous to our method in the θ faces, we appeal to the Cartesian definition of $\mathbf{e}_\phi(\phi)$

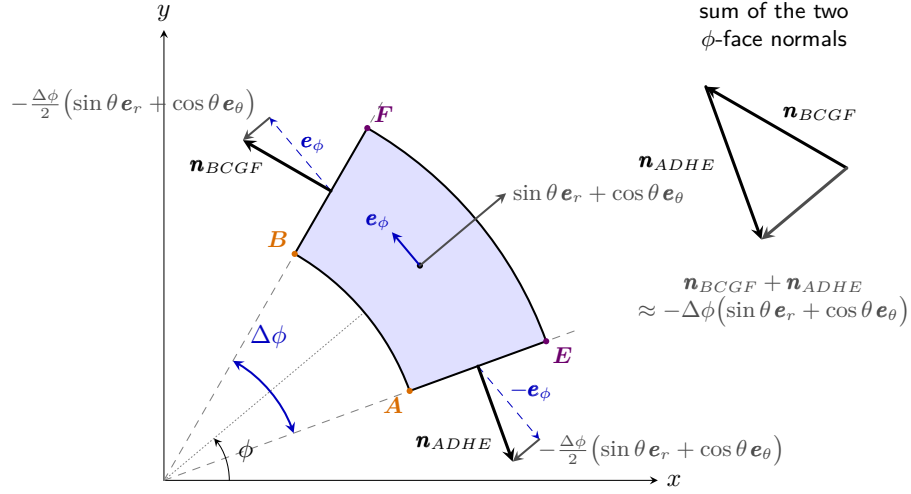
$$\mathbf{e}_\phi(\phi) = -\sin(\phi) \hat{\mathbf{x}} + \cos(\phi) \hat{\mathbf{y}}$$

and Taylor expand $\mathbf{e}_\phi(\phi \pm \Delta\phi/2)$ to find

$$\begin{aligned} \mathbf{e}_\phi\left(\phi \pm \frac{\Delta\phi}{2}\right) &= \mathbf{e}_\phi(\phi) \pm \frac{\Delta\phi}{2} \frac{\partial \mathbf{e}_\phi}{\partial \phi} \\ &= \mathbf{e}_\phi(\phi) \mp \frac{\Delta\phi}{2} [\sin(\theta) \mathbf{e}_r + \cos(\theta) \mathbf{e}_\theta] \end{aligned}$$

which we see agrees with the geometry in the figure below. The normals along the aforementioned faces are thus

$$\begin{aligned} [\mathbf{n} dA]_{ADHE} &= -\left[\mathbf{e}_\phi + \mathbf{e}_r \frac{\Delta\phi}{2} \sin(\theta) + \mathbf{e}_\theta \frac{\Delta\phi}{2} \cos(\theta) \right] r \Delta r \Delta \theta \\ [\mathbf{n} dA]_{BCGF} &= \left[\mathbf{e}_\phi - \mathbf{e}_r \frac{\Delta\phi}{2} \sin(\theta) - \mathbf{e}_\theta \frac{\Delta\phi}{2} \cos(\theta) \right] r \Delta r \Delta \theta \end{aligned}$$



The final step is to find the Taylor expansions of $\mathbf{Q}(r, \theta, \phi)$ at each of the six faces, and compute the dot product with the normals. In general, for a variable α , we have

$$\mathbf{Q}\left(\alpha \pm \frac{\Delta\alpha}{2}\right) = \mathbf{Q}(\alpha) \pm \frac{\partial \mathbf{Q}}{\partial \alpha} \frac{\Delta\alpha}{2}$$

Moreover, from the component definition of $\mathbf{Q}(r, \theta, \phi)$

$$\mathbf{Q}(r, \theta, \phi) = \mathbf{e}_r Q_r + \mathbf{e}_\theta Q_\theta + \mathbf{e}_\phi Q_\phi$$

we compute derivatives

$$\begin{aligned} \frac{\partial \mathbf{Q}}{\partial r} &= \mathbf{e}_r \frac{\partial Q_r}{\partial r} + \mathbf{e}_\theta \frac{\partial Q_\theta}{\partial r} + \mathbf{e}_\phi \frac{\partial Q_\phi}{\partial r} \\ \frac{\partial \mathbf{Q}}{\partial \theta} &= \left(\mathbf{e}_r \frac{\partial Q_r}{\partial \theta} + \frac{\partial \mathbf{e}_r}{\partial \theta} Q_r \right) + \left(\mathbf{e}_\theta \frac{\partial Q_\theta}{\partial \theta} + \frac{\partial \mathbf{e}_\theta}{\partial \theta} Q_\theta \right) + \mathbf{e}_\phi \frac{\partial Q_\phi}{\partial \theta} \\ &= \mathbf{e}_r \left(\frac{\partial Q_r}{\partial \theta} - Q_\theta \right) + \mathbf{e}_\theta \left(\frac{\partial Q_\theta}{\partial \theta} + Q_r \right) + \mathbf{e}_\phi \frac{\partial Q_\phi}{\partial \theta} \\ \frac{\partial \mathbf{Q}}{\partial \phi} &= \left(\mathbf{e}_r \frac{\partial Q_r}{\partial \phi} + \frac{\partial \mathbf{e}_r}{\partial \phi} Q_r \right) + \left(\mathbf{e}_\theta \frac{\partial Q_\theta}{\partial \phi} + \frac{\partial \mathbf{e}_\theta}{\partial \phi} Q_\theta \right) + \left(\mathbf{e}_\phi \frac{\partial Q_\phi}{\partial \phi} + \frac{\partial \mathbf{e}_\phi}{\partial \phi} Q_\phi \right) \\ &= \mathbf{e}_r \left(\frac{\partial Q_r}{\partial \phi} - Q_\phi \sin(\theta) \right) + \mathbf{e}_\theta \left(\frac{\partial Q_\theta}{\partial \phi} - Q_\phi \cos(\theta) \right) \\ &\quad + \mathbf{e}_\phi \left(\frac{\partial Q_\phi}{\partial \phi} + Q_r \sin(\theta) + Q_\theta \cos(\theta) \right) \end{aligned}$$

We finally have what we need for the penultimate step of computing dot products:

$$\begin{aligned}
[\mathbf{n} \cdot \mathbf{Q} dA]_{ABCD} &= \left[\mathbf{Q} - \frac{\partial \mathbf{Q}}{\partial r} \frac{\Delta r}{2} \right] \cdot [-\mathbf{e}_r (r^2 - r \Delta r) \sin(\theta) \Delta \theta \Delta \phi] \\
&= -Q_r (r^2 - r \Delta r) \sin(\theta) \Delta \theta \Delta \phi \\
&\quad + \frac{\partial Q_r}{\partial r} \frac{\Delta r}{2} (r^2 - r \Delta r) \sin(\theta) \Delta \theta \Delta \phi \\
&= -Q_r (r^2 - r \Delta r) \sin(\theta) \Delta \theta \Delta \phi \\
&\quad + \frac{1}{2} \frac{\partial Q_r}{\partial r} r^2 \sin(\theta) \Delta r \Delta \theta \Delta \phi \\
[\mathbf{n} \cdot \mathbf{Q} dA]_{EFGH} &= \left(\mathbf{Q} + \frac{\partial \mathbf{Q}}{\partial r} \frac{\Delta r}{2} \right) \cdot \mathbf{e}_r (r^2 + r \Delta r) \sin(\theta) \Delta \theta \Delta \phi \\
&= Q_r (r^2 + r \Delta r) \sin(\theta) \Delta \theta \Delta \phi \\
&\quad + \frac{\partial Q_r}{\partial r} \frac{\Delta r}{2} (r^2 + r \Delta r) \sin(\theta) \Delta \theta \Delta \phi \\
&= Q_r (r^2 + r \Delta r) \sin(\theta) \Delta \theta \Delta \phi \\
&\quad + \frac{1}{2} \frac{\partial Q_r}{\partial r} r^2 \sin(\theta) \Delta r \Delta \theta \Delta \phi \\
[\mathbf{n} \cdot \mathbf{Q} dA]_{ABCD} + [\mathbf{n} \cdot \mathbf{Q} dA]_{EFGH} &= \boxed{\left(\frac{2Q_r}{r} + \frac{\partial Q_r}{\partial r} \right) r^2 \sin(\theta) \Delta r \Delta \theta \Delta \phi} \\
[\mathbf{n} \cdot \mathbf{Q} dA]_{ABFE} &= \left[\mathbf{Q} - \frac{\partial \mathbf{Q}}{\partial \theta} \frac{\Delta \theta}{2} \right] \cdot \left[\mathbf{e}_\theta \left[-r \sin(\theta) + \frac{r \Delta \theta}{2} \cos(\theta) \right] \Delta r \Delta \phi - \mathbf{e}_r \frac{r \Delta \theta}{2} \sin(\theta) \Delta r \Delta \phi \right] \\
&= Q_\theta \left[-r \sin(\theta) + \frac{r \Delta \theta}{2} \cos(\theta) \right] \Delta r \Delta \phi - Q_r \frac{r \Delta \theta}{2} \sin(\theta) \Delta r \Delta \phi \\
&\quad + \left(\frac{\partial Q_\theta}{\partial \theta} + Q_r \right) \frac{\Delta \theta}{2} \left[r \sin(\theta) - \frac{r \Delta \theta}{2} \cos(\theta) \right] \Delta r \Delta \phi \\
&\quad + \left(\frac{\partial Q_r}{\partial \theta} - Q_\theta \right) \frac{r (\Delta \theta)^2}{4} \sin(\theta) \Delta r \Delta \phi \\
&= -Q_\theta r \sin(\theta) \Delta r \Delta \phi + Q_\theta \frac{r \Delta \theta}{2} \cos(\theta) \Delta r \Delta \phi + \frac{1}{2r} \frac{\partial Q_\theta}{\partial \theta} r^2 \sin(\theta) \Delta r \Delta \theta \Delta \phi \\
[\mathbf{n} \cdot \mathbf{Q} dA]_{CDHG} &= \left[\mathbf{Q} + \frac{\partial \mathbf{Q}}{\partial \theta} \frac{\Delta \theta}{2} \right] \cdot \left[\mathbf{e}_\theta \left[r \sin(\theta) + \frac{r \Delta \theta}{2} \cos(\theta) \right] \Delta r \Delta \phi - \mathbf{e}_r \frac{r \Delta \theta}{2} \sin(\theta) \Delta r \Delta \phi \right] \\
&= Q_\theta \left[r \sin(\theta) + \frac{r \Delta \theta}{2} \cos(\theta) \right] \Delta r \Delta \phi - Q_r \frac{r \Delta \theta}{2} \sin(\theta) \Delta r \Delta \phi \\
&\quad + \left(\frac{\partial Q_\theta}{\partial \theta} + Q_r \right) \frac{\Delta \theta}{2} \left[r \sin(\theta) + \frac{r \Delta \theta}{2} \cos(\theta) \right] \Delta r \Delta \phi \\
&\quad - \left(\frac{\partial Q_r}{\partial \theta} - Q_\theta \right) \frac{r (\Delta \theta)^2}{4} \sin(\theta) \Delta r \Delta \phi \\
&= Q_\theta r \sin(\theta) \Delta r \Delta \phi + Q_\theta \frac{r \Delta \theta}{2} \cos(\theta) \Delta r \Delta \phi + \frac{1}{2r} \frac{\partial Q_\theta}{\partial \theta} r^2 \sin(\theta) \Delta r \Delta \theta \Delta \phi \\
[\mathbf{n} \cdot \mathbf{Q} dA]_{ABFE} + [\mathbf{n} \cdot \mathbf{Q} dA]_{CDHG} &= \boxed{\left(\frac{Q_\theta}{r} \cot(\theta) + \frac{1}{r} \frac{\partial Q_\theta}{\partial \theta} \right) r^2 \sin(\theta) \Delta r \Delta \theta \Delta \phi}
\end{aligned}$$

$$\begin{aligned}
[\mathbf{n} \cdot \mathbf{Q} dA]_{ADHE} &= \left[\mathbf{Q} - \frac{\partial \mathbf{Q}}{\partial \phi} \frac{\Delta \phi}{2} \right] \cdot \left[- \left[\mathbf{e}_\phi + \mathbf{e}_r \frac{\Delta \phi}{2} \sin(\theta) + \mathbf{e}_\theta \frac{\Delta \phi}{2} \cos(\theta) \right] r \Delta r \Delta \theta \right] \\
&= - \left(Q_r \frac{\Delta \phi}{2} \sin(\theta) + Q_\theta \frac{\Delta \phi}{2} \cos(\theta) + Q_\phi \right) r \Delta r \Delta \theta \\
&\quad + \left[\left(\frac{\partial Q_r}{\partial \phi} + Q_\phi \sin(\theta) \right) \left(\frac{\Delta \phi}{2} \right)^2 \sin(\theta) + \left(\frac{\partial Q_\theta}{\partial \phi} + Q_\phi \cos(\theta) \right) \left(\frac{\Delta \phi}{2} \right)^2 \cos(\theta) \right. \\
&\quad \left. + \left(\frac{\partial Q_\phi}{\partial \phi} + Q_r \sin(\theta) + Q_\theta \cos(\theta) \right) \left(\frac{\Delta \phi}{2} \right) \right] r \Delta r \Delta \theta \\
&= -Q_\phi r \Delta r \Delta \theta + \frac{1}{2} \frac{\partial Q_\phi}{\partial \phi} r \Delta r \Delta \theta \Delta \phi \\
[\mathbf{n} \cdot \mathbf{Q} dA]_{BCGF} &= \left[\mathbf{Q} + \frac{\partial \mathbf{Q}}{\partial \phi} \frac{\Delta \phi}{2} \right] \cdot \left[\mathbf{e}_\phi - \mathbf{e}_r \frac{\Delta \phi}{2} \sin(\theta) - \mathbf{e}_\theta \frac{\Delta \phi}{2} \cos(\theta) \right] r \Delta r \Delta \theta \\
&= \left(-Q_r \frac{\Delta \phi}{2} \sin(\theta) - Q_\theta \frac{\Delta \phi}{2} \cos(\theta) + Q_\phi \right) r \Delta r \Delta \theta \\
&\quad + \left[- \left(\frac{\partial Q_r}{\partial \phi} + Q_\phi \sin(\theta) \right) \left(\frac{\Delta \phi}{2} \right)^2 \sin(\theta) - \left(\frac{\partial Q_\theta}{\partial \phi} + Q_\phi \cos(\theta) \right) \left(\frac{\Delta \phi}{2} \right)^2 \cos(\theta) \right. \\
&\quad \left. + \left(\frac{\partial Q_\phi}{\partial \phi} + Q_r \sin(\theta) + Q_\theta \cos(\theta) \right) \left(\frac{\Delta \phi}{2} \right) \right] r \Delta r \Delta \theta \\
&= Q_\phi r \Delta r \Delta \theta + \frac{1}{2} \frac{\partial Q_\phi}{\partial \phi} r \Delta r \Delta \theta \Delta \phi \\
[\mathbf{n} \cdot \mathbf{Q} dA]_{ADHE} + [\mathbf{n} \cdot \mathbf{Q} dA]_{BCGF} &= \boxed{\frac{1}{r \sin(\theta)} \frac{\partial Q_\phi}{\partial \phi} r^2 \Delta r \sin(\theta) \Delta \theta \Delta \phi}
\end{aligned}$$

At last, summing the terms produces

$$\begin{aligned}
\nabla \cdot \mathbf{Q} &= \lim_{V \rightarrow 0} \frac{1}{V} \iint_A \mathbf{n} \cdot \mathbf{Q} dA \\
&= \lim_{\substack{\Delta r \rightarrow 0 \\ \Delta \theta \rightarrow 0 \\ \Delta \phi \rightarrow 0}} \frac{1}{r^2 \sin(\theta) \Delta r \Delta \theta \Delta \phi} \left(\frac{2Q_r}{r} + \frac{\partial Q_r}{\partial r} + \frac{1}{r} \frac{\partial Q_\theta}{\partial \theta} + \frac{\cot(\theta)}{r} Q_\theta + \frac{1}{r \sin(\theta)} \frac{\partial Q_\phi}{\partial \phi} \right) r^2 \sin(\theta) \Delta r \Delta \theta \Delta \phi \\
&= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 Q_r) + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \theta} (\sin(\theta) Q_\theta) + \frac{1}{r \sin(\theta)} \frac{\partial Q_\phi}{\partial \phi}
\end{aligned}$$

Question: 2.21

Use the vector integral theorems to prove that $\nabla \cdot (\nabla \times \mathbf{u}) = 0$ for any twice-differentiable vector function $\mathbf{u}$ regardless of the coordinate system.

By Gauss's theorem,

$$\iiint_V \nabla \cdot (\nabla \times \mathbf{u}) dV = \iint_A \mathbf{n} \cdot (\nabla \times \mathbf{u}) dA$$

Now, intuition tells us that Stokes' theorem is next, which is correct. Some nuance, however, is required. Stokes' theorem deals with an *open* surface area and the bounding curve around the surface. With a closed surface, we can take an arbitrary cut through it, fix a direction, and call that our boundary curve C . This produces

two *open* surfaces. Because the direction of that curve is fixed, the right-hand rule traverses the curve C in opposite directions on each surface and their line integrals cancel, yielding zero under Stokes' theorem.

$$\iint_A \mathbf{n} \cdot (\nabla \times \mathbf{u}) \, dA = \oint_C \mathbf{u} \cdot d\mathbf{l} - \oint_C \mathbf{u} \cdot d\mathbf{l} = 0$$

Lastly, since V can be any arbitrary volume, by continuity of the integrand as established by the twice-differentiability of $\mathbf{u}$, we establish $\nabla \cdot (\nabla \times \mathbf{u}) = 0$.

3 - Kinematics

Question: 3.1

The gradient operator in Cartesian coordinates (x, y, z) is: $\nabla = \mathbf{e}_x(\partial/\partial x) + \mathbf{e}_y(\partial/\partial y) + \mathbf{e}_z(\partial/\partial z)$ where $\mathbf{e}_x$, $\mathbf{e}_y$, and $\mathbf{e}_z$ are the unit vectors. In cylindrical polar coordinates (R, ϕ, z) having the same origin, (see Figure 3.3c), coordinates and unit vectors are related by: $R = \sqrt{x^2 + y^2}$, $\phi = \tan^{-1}(y/x)$, and $z = z$; and $\mathbf{e}_R = \mathbf{e}_x \cos(\phi) + \mathbf{e}_y \sin(\phi)$, $\mathbf{e}_\phi = -\mathbf{e}_x \sin(\phi) + \mathbf{e}_y \cos(\phi)$, and $\mathbf{e}_z = \mathbf{e}_z$. Determine the following in the cylindrical polar coordinate system. [This exercise requires some mathematical patience.]

- (c) the divergence of the velocity field $\nabla \cdot \mathbf{u}$
- (e) the advective acceleration term $(\mathbf{u} \cdot \nabla) \mathbf{u}$

- (c) First, the definitions for R and ϕ collectively produce

$$x = R \cos(\phi) \quad y = R \sin(\phi)$$

Given the gradient operator in Cartesian coordinates, we will convert the Cartesian basis vectors and differential operators into their counterparts in cylindrical polar coordinates. We begin with the basis vector conversions. First we find the projections of $\mathbf{e}_x$ and $\mathbf{e}_y$ on $\mathbf{e}_R$ and $\mathbf{e}_\phi$; since $\mathbf{e}_z$ is unchanged we already know orthogonality exists.

$$\begin{aligned} \mathbf{e}_x \cdot \mathbf{e}_R &= \cos(\phi) \\ \mathbf{e}_x \cdot \mathbf{e}_\phi &= -\sin(\phi) \\ \mathbf{e}_y \cdot \mathbf{e}_R &= \sin(\phi) \\ \mathbf{e}_y \cdot \mathbf{e}_\phi &= \cos(\phi) \end{aligned}$$

Expressing $\mathbf{e}_x$ and $\mathbf{e}_y$ in cylindrical polar coordinates:

$$\begin{aligned} \mathbf{e}_x &= \cos(\phi) \mathbf{e}_R - \sin(\phi) \mathbf{e}_\phi \\ \mathbf{e}_y &= \sin(\phi) \mathbf{e}_R + \cos(\phi) \mathbf{e}_\phi \end{aligned}$$

which we also could have derived by identifying the corresponding linear combinations of the R and ϕ basis vectors.

Following this, we convert the differential operators. Assisting in this endeavor is the total derivative:

$$\begin{aligned} \frac{\partial}{\partial x} &= \frac{\partial R}{\partial x} \frac{\partial}{\partial R} + \frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi} \\ &= \cos(\phi) \frac{\partial}{\partial R} - \frac{\sin(\phi)}{R} \frac{\partial}{\partial \phi} \\ \frac{\partial}{\partial y} &= \frac{\partial R}{\partial y} \frac{\partial}{\partial R} + \frac{\partial \phi}{\partial y} \frac{\partial}{\partial \phi} \\ &= \sin(\phi) \frac{\partial}{\partial R} + \frac{\cos(\phi)}{R} \frac{\partial}{\partial \phi} \end{aligned}$$

where the partial derivatives can be derived from the definitions of R and ϕ . Substituting these pieces into the Cartesian gradient operator gives us

$$\begin{aligned} & (\cos(\phi) \mathbf{e}_R - \sin(\phi) \mathbf{e}_\theta) \left(\cos(\phi) \frac{\partial}{\partial R} - \frac{\sin(\phi)}{R} \frac{\partial}{\partial \phi} \right) \\ & + (\sin(\phi) \mathbf{e}_R + \cos(\phi) \mathbf{e}_\theta) \left(\sin(\phi) \frac{\partial}{\partial R} + \frac{\cos(\phi)}{R} \frac{\partial}{\partial \phi} \right) + \mathbf{e}_z \frac{\partial}{\partial z} \\ & = \mathbf{e}_R \frac{\partial}{\partial R} + \mathbf{e}_\phi \frac{1}{R} \frac{\partial}{\partial \phi} + \mathbf{e}_z \frac{\partial}{\partial z} \end{aligned}$$

For $\mathbf{u} = \mathbf{e}_R u_R + \mathbf{e}_\phi u_\phi + \mathbf{e}_z u_z$,

$$\nabla \cdot \mathbf{u} = \frac{\partial u_R}{\partial R} + \frac{1}{R} \frac{\partial u_\phi}{\partial \phi} + \frac{\partial u_z}{\partial z}$$

(e) Compute

$$\mathbf{u} \cdot \nabla = u_R \frac{\partial}{\partial R} + \frac{u_\phi}{R} \frac{\partial}{\partial \phi} + u_z \frac{\partial}{\partial z}$$

For the next step, we will need to apply each component of $\mathbf{u}$ with our derived operator. Some facts to know that will assist with our computation are that

$$\frac{\partial \mathbf{e}_R}{\partial \phi} = \mathbf{e}_\phi \quad \frac{\partial \mathbf{e}_\phi}{\partial \phi} = -\mathbf{e}_R$$

are the only non-zero derivatives of any cylindrical polar basis vectors with respect to any variable. Each component is derived as

$$\begin{aligned} (\mathbf{u} \cdot \nabla)(\mathbf{e}_R u_R) &= \mathbf{e}_R \left(u_R \frac{\partial u_R}{\partial R} + \frac{u_\phi}{R} \frac{\partial u_R}{\partial \phi} + u_z \frac{\partial u_R}{\partial z} \right) + \mathbf{e}_\phi \frac{u_R u_\phi}{R} \\ (\mathbf{u} \cdot \nabla)(\mathbf{e}_\phi u_\phi) &= -\mathbf{e}_R \frac{u_\phi^2}{R} + \mathbf{e}_\phi \left(u_R \frac{\partial u_\phi}{\partial R} + \frac{u_\phi}{R} \frac{\partial u_\phi}{\partial \phi} + u_z \frac{\partial u_\phi}{\partial z} \right) \\ (\mathbf{u} \cdot \nabla)(\mathbf{e}_z u_z) &= \mathbf{e}_z \left(u_R \frac{\partial u_z}{\partial R} + \frac{u_\phi}{R} \frac{\partial u_z}{\partial \phi} + u_z \frac{\partial u_z}{\partial z} \right) \end{aligned}$$

Summing, conclude

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = \mathbf{e}_R \left[(\mathbf{u} \cdot \nabla) u_R - \frac{u_\phi^2}{R} \right] + \mathbf{e}_\phi \left[(\mathbf{u} \cdot \nabla) u_\phi + \frac{u_R u_\phi}{R} \right] + \mathbf{e}_z (\mathbf{u} \cdot \nabla) u_z$$

Observe that the u_ϕ^2/R is the centripetal acceleration term as it points inward towards the origin along the R -axis. The second of the unusual terms, $u_R u_\phi/R$, can be thought of as the part of the radial velocity that is directed azimuthally as the fluid parcel moves. Conceptually, this is an angular momentum conservation term, which we will explore more in later chapters.

Question: 3.2

Consider Cartesian coordinates (as given in Exercise 3.1) and spherical polar coordinates (r, θ, ϕ) having the same origin (see Figure 3.3d). Here coordinates and unit vectors are related by: $r = \sqrt{x^2 + y^2 + z^2}$, $\theta = \tan^{-1}(\sqrt{x^2 + y^2}/z)$, and $\phi = \tan^{-1}(y/x)$; and $\mathbf{e}_r = \mathbf{e}_x \cos(\phi) \sin(\theta) + \mathbf{e}_y \sin(\phi) \sin(\theta) + \mathbf{e}_z \cos(\theta)$, $\mathbf{e}_\theta = \mathbf{e}_x \cos(\phi) \cos(\theta) + \mathbf{e}_y \sin(\phi) \cos(\theta) - \mathbf{e}_z \sin(\theta)$, and $\mathbf{e}_\phi = -\mathbf{e}_x \sin(\phi) + \mathbf{e}_y \cos(\phi)$. In the spherical polar coordinate system, determine the following items. [This exercise requires mathematical patience.]

- (a) $\partial \mathbf{e}_r / \partial \theta$, $\partial \mathbf{e}_r / \partial \phi$, $\partial \mathbf{e}_\theta / \partial \theta$, $\partial \mathbf{e}_\theta / \partial \phi$, and $\partial \mathbf{e}_\phi / \partial \phi$
- (b) the gradient operator ∇
- (c) the divergence of the velocity field $\nabla \cdot \mathbf{u}$
- (d) the Laplacian operator $\nabla \cdot \nabla \equiv \nabla^2$
- (e) the advective acceleration term $(\mathbf{u} \cdot \nabla) \mathbf{u}$

(a) Compute

$$\begin{aligned}
 \frac{\partial \mathbf{e}_r}{\partial \theta} &= \mathbf{e}_x \cos(\phi) \cos(\theta) + \mathbf{e}_y \sin(\phi) \cos(\theta) - \mathbf{e}_z \sin(\theta) \\
 &= \mathbf{e}_\theta \\
 \frac{\partial \mathbf{e}_r}{\partial \phi} &= -\mathbf{e}_x \sin(\phi) \sin(\theta) + \mathbf{e}_y \cos(\phi) \sin(\theta) \\
 &= \mathbf{e}_\phi \sin(\theta) \\
 \frac{\partial \mathbf{e}_\theta}{\partial \theta} &= -\mathbf{e}_x \cos(\phi) \sin(\theta) - \mathbf{e}_y \sin(\phi) \sin(\theta) - \mathbf{e}_z \cos(\theta) \\
 &= -\mathbf{e}_r \\
 \frac{\partial \mathbf{e}_\theta}{\partial \phi} &= -\mathbf{e}_x \sin(\phi) \cos(\theta) + \mathbf{e}_y \cos(\phi) \cos(\theta) \\
 &= \mathbf{e}_\phi \cos(\theta) \\
 \frac{\partial \mathbf{e}_\phi}{\partial \phi} &= -\mathbf{e}_x \cos(\phi) - \mathbf{e}_y \sin(\phi) \\
 &= -\mathbf{e}_r \sin(\theta) - \mathbf{e}_\theta \cos(\theta)
 \end{aligned}$$

(b) There is a brute force way and an elegant way. We will do it the painful way first so as to appreciate the elegance of a more geometric argument.

First we write each Cartesian basis vector in terms of their spherical polar counterparts. A slick way to do this is to recognize that in matrix form

$$\underbrace{\begin{bmatrix} \cos(\phi) \sin(\theta) & \sin(\phi) \sin(\theta) & \cos(\theta) \\ \cos(\phi) \cos(\theta) & \sin(\phi) \cos(\theta) & -\sin(\theta) \\ -\sin(\phi) & \cos(\phi) & 0 \end{bmatrix}}_{\mathbf{Q}} \begin{bmatrix} \mathbf{e}_x \\ \mathbf{e}_y \\ \mathbf{e}_z \end{bmatrix} = \begin{bmatrix} \mathbf{e}_r \\ \mathbf{e}_\theta \\ \mathbf{e}_\phi \end{bmatrix}$$

we can invert the orthonormal basis by virtue of the fact that $\mathbf{Q}^T = \mathbf{Q}^{-1}$, and derive

$$\begin{aligned}\mathbf{e}_x &= \mathbf{e}_r \cos(\phi) \sin(\theta) + \mathbf{e}_\theta \cos(\phi) \cos(\theta) - \mathbf{e}_\phi \sin(\phi) \\ \mathbf{e}_y &= \mathbf{e}_r \sin(\phi) \sin(\theta) + \mathbf{e}_\theta \sin(\phi) \cos(\theta) + \mathbf{e}_\phi \cos(\phi) \\ \mathbf{e}_z &= \mathbf{e}_r \cos(\theta) - \mathbf{e}_\theta \sin(\theta)\end{aligned}$$

Next we do the conversion of differential operators from Cartesian to spherical polar. To do so, begin with the conversions between Cartesian to spherical polar variables:

$$x = r \sin(\theta) \cos(\phi), \quad y = r \sin(\theta) \sin(\phi), \quad z = r \cos(\theta)$$

Then using the definitions of r , θ , and ϕ , compute (using the Cartesian gradient)

$$\begin{aligned}\nabla r &= \left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r} \right) = (\cos(\phi) \sin(\theta), \sin(\phi) \sin(\theta), \cos(\theta)) = \mathbf{e}_r \\ \nabla \theta &= \frac{1}{r} (\cos(\phi) \cos(\theta), \sin(\phi) \cos(\theta), -\sin(\theta)) = \frac{\mathbf{e}_\theta}{r} \\ \nabla \phi &= \frac{1}{r \sin(\theta)} (-\sin(\phi), \cos(\phi), 0) = \frac{\mathbf{e}_\phi}{r \sin(\theta)}\end{aligned}$$

Using the total derivative, we can then derive

$$\begin{aligned}\frac{\partial}{\partial x} &= \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} + \frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi} \\ &= \cos(\phi) \sin(\theta) \frac{\partial}{\partial r} + \frac{\cos(\phi) \cos(\theta)}{r} \frac{\partial}{\partial \theta} - \frac{\sin(\phi)}{r \sin(\theta)} \frac{\partial}{\partial \phi} \\ \frac{\partial}{\partial y} &= \frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial}{\partial \theta} + \frac{\partial \phi}{\partial y} \frac{\partial}{\partial \phi} \\ &= \sin(\phi) \sin(\theta) \frac{\partial}{\partial r} + \frac{\sin(\phi) \cos(\theta)}{r} \frac{\partial}{\partial \theta} + \frac{\cos(\phi)}{r \sin(\theta)} \frac{\partial}{\partial \phi} \\ \frac{\partial}{\partial z} &= \frac{\partial r}{\partial z} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial z} \frac{\partial}{\partial \theta} + \frac{\partial \phi}{\partial z} \frac{\partial}{\partial \phi} \\ &= \cos(\theta) \frac{\partial}{\partial r} - \frac{\sin(\theta)}{r} \frac{\partial}{\partial \theta}\end{aligned}$$

Manually substituting these differential operators with the basis vectors we derived into the Cartesian gradient is gruelling. To abbreviate our work, we will make use of Einstein summation. To do so, make the convenient observation that the partial derivatives used to derive the operators (sans the scaling coefficients) and components of the basis vectors are the **columns** of $\mathbf{Q}$. There is a common structure in our work that originates from the orthonormal matrix $\mathbf{Q}$ as we have color-coded above.

We now derive the summation formula for the gradient operator. Let $q_j = (r, \theta, \phi)$, $x_i = (x, y, z)$, and $h_j = (1, r, r \sin(\theta))$. Then we have

$$\nabla = \underbrace{\mathbf{e}_{q_j} h_j}_{\mathbf{e}_{x_i}} \underbrace{\frac{\partial q_j}{\partial x_i} \frac{\partial q_k}{\partial x_i} \frac{\partial}{\partial q_k}}_{\partial / \partial x_i}$$

Now, observe that for the j -th row and i -th column of Q

$$Q_{ji} = h_j \frac{\partial q_j}{\partial x_i}$$

Substituting into our summation produces

$$\nabla = \mathbf{e}_{q_j} \frac{1}{h_k} Q_{ji} Q_{ki} \frac{\partial}{\partial q_k}$$

And now for the finale: the Q 's are elements of an **orthonormal** matrix. The beauty of this structure is we can definitively say $Q_{ji} Q_{ki} = Q_{ji} Q_{ik}^T = \delta_{jk}$. Thus we derive the gradient operator in spherical coordinates:

$$\nabla = \mathbf{e}_{q_j} \frac{1}{h_j} \frac{\partial}{\partial q_j} = \mathbf{e}_r \frac{\partial}{\partial r} + \mathbf{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \mathbf{e}_\phi \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \phi}$$

Each of the individual h_j 's are **scaling factors** that are uniquely derived for each coordinate basis. Crucially, the *structure* of the gradient operator is consistent across *all* **orthogonal** coordinate bases.

The elegant way. For a scalar function f , the coordinate-free definition of the gradient is

$$df = \nabla f \cdot d\mathbf{s}$$

where we have the infinitesimal displacement

$$d\mathbf{s} = \mathbf{e}_r dr + \mathbf{e}_\theta r d\theta + \mathbf{e}_\phi r \sin(\theta) d\phi \equiv \mathbf{e}_{q_j} h_j dq_j$$

and by the chain rule

$$df = \frac{\partial f}{\partial r} dr + \frac{\partial f}{\partial \theta} d\theta + \frac{\partial f}{\partial \phi} d\phi \equiv \frac{\partial f}{\partial q_j} dq_j$$

Next, define ∇f as

$$\nabla f = \mathbf{e}_r F_r + \mathbf{e}_\theta F_\theta + \mathbf{e}_\phi F_\phi \equiv \mathbf{e}_{q_j} F_j$$

By orthogonality, $\mathbf{e}_{q_j} \cdot \mathbf{e}_{q_k} = \delta_{jk}$. Dotting ∇f with $d\mathbf{s}$ yields

$$F_r dr + F_\theta r d\theta + F_\phi r \sin(\theta) d\phi$$

which allows us to deduce

$$F_r = \frac{\partial f}{\partial r}, \quad F_\theta = \frac{1}{r} \frac{\partial f}{\partial \theta}, \quad F_\phi = \frac{1}{r \sin(\theta)} \frac{\partial f}{\partial \phi}$$

which are precisely the components to the gradient operator in spherical polar coordinates. Much easier!

(c) Define $\mathbf{u} = \mathbf{e}_r u_r + \mathbf{e}_\theta u_\theta + \mathbf{e}_\phi u_\phi \equiv \mathbf{e}_{q_j} u_{q_j}$. Compute

$$\begin{aligned} \nabla \cdot \mathbf{u} &= \left(\mathbf{e}_{q_j} \frac{1}{h_j} \frac{\partial}{\partial q_j} \right) \cdot (\mathbf{e}_{q_k} u_{q_k}) \\ &= \mathbf{e}_{q_j} \cdot \frac{\partial \mathbf{e}_{q_k}}{\partial q_j} \frac{1}{h_j} u_{q_k} + \underbrace{\mathbf{e}_{q_j} \cdot \mathbf{e}_{q_k}}_{\delta_{jk}} \frac{1}{h_j} \frac{\partial u_{q_k}}{\partial q_j} \\ &= \mathbf{e}_{q_j} \cdot \frac{\partial \mathbf{e}_{q_k}}{\partial q_j} \frac{1}{h_j} u_{q_k} + \frac{1}{h_j} \frac{\partial u_{q_j}}{\partial q_j} \end{aligned}$$

Using the results from part (a), we must manually evaluate the first term. First observe that if $j = k$, by normalization and orthogonality of a unit vector and its derivative, $\mathbf{e}_{q_j} \cdot \partial \mathbf{e}_{q_j} / \partial q_j = 0$. Additionally, when $q_j = r$, since no unit vector is a function of r , the term also vanishes. This leaves only three non-zero options: $\mathbf{e}_\theta \cdot \partial \mathbf{e}_r / \partial \theta = 1$, $\mathbf{e}_\phi \cdot \partial \mathbf{e}_r / \partial \phi = \sin(\theta)$, and $\mathbf{e}_\phi \cdot \partial \mathbf{e}_\theta / \partial \phi = \cos(\theta)$. (Note that $\mathbf{e}_\phi$ does not depend on θ .) Those terms produce

$$\frac{1}{r} u_r + \frac{\sin(\theta)}{r \sin(\theta)} u_r + \frac{\cos(\theta)}{r \sin(\theta)} u_\theta$$

Conclude

$$\begin{aligned} \nabla \cdot \mathbf{u} &= \frac{2}{r} u_r + \frac{\cot(\theta)}{r} u_\theta + \frac{\partial u_r}{\partial r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{1}{r \sin(\theta)} \frac{\partial u_\phi}{\partial \phi} \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u_r) + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \theta} (\sin(\theta) u_\theta) + \frac{1}{r \sin(\theta)} \frac{\partial u_\phi}{\partial \phi} \end{aligned}$$

(d) Compute

$$\begin{aligned} \nabla \cdot \nabla &\equiv \nabla^2 = \left(\mathbf{e}_{q_j} \frac{1}{h_j} \frac{\partial}{\partial q_j} \right) \cdot \left(\mathbf{e}_{q_k} \frac{1}{h_k} \frac{\partial}{\partial q_k} \right) \\ &= \mathbf{e}_{q_j} \cdot \frac{\partial \mathbf{e}_{q_k}}{\partial q_j} \left(\frac{1}{h_j h_k} \frac{\partial}{\partial q_k} \right) + \underbrace{\mathbf{e}_{q_j} \cdot \mathbf{e}_{q_k}}_{\delta_{jk}} \left(\frac{1}{h_j} \frac{\partial(1/h_k)}{\partial q_j} \frac{\partial}{\partial q_k} \right) \\ &\quad + \underbrace{\mathbf{e}_{q_j} \cdot \mathbf{e}_{q_k}}_{\delta_{jk}} \frac{1}{h_j h_k} \frac{\partial^2}{\partial q_j \partial q_k} \end{aligned}$$

Analogously to part (c), the first term is non-zero for only three (j, k) pairs. The second term requires $j = k$ for orthonormality of the unit vectors, but in doing so, $\partial(1/h_k) / \partial q_j$ vanishes for each individual pair (h_j, q_j) , as each scale factor is independent of its coordinate. The third term is non-zero only when $j = k$. Evaluate each term:

$$\begin{aligned} \mathbf{e}_{q_j} \cdot \frac{\partial \mathbf{e}_{q_k}}{\partial q_j} \left(\frac{1}{h_j h_k} \frac{\partial}{\partial q_k} \right) &= \frac{1}{r} \frac{\partial}{\partial r} + \frac{\sin(\theta)}{r \sin(\theta)} \frac{\partial}{\partial r} + \frac{\cot(\theta)}{r^2} \frac{\partial}{\partial \theta} \\ &= \frac{2}{r} \frac{\partial}{\partial r} + \frac{\cot(\theta)}{r^2} \frac{\partial}{\partial \theta} \\ \mathbf{e}_{q_j} \cdot \mathbf{e}_{q_k} \left(\frac{1}{h_j} \frac{\partial(1/h_k)}{\partial q_j} \frac{\partial}{\partial q_k} \right) &= \frac{1}{h_j} \frac{\partial(1/h_j)}{\partial q_j} \frac{\partial}{\partial q_j} \\ &= -\frac{1}{h_j^3} \frac{\partial h_j}{\partial q_j} \frac{\partial}{\partial q_j} \\ &= 0 \\ \mathbf{e}_{q_j} \cdot \mathbf{e}_{q_k} \frac{1}{h_j h_k} \frac{\partial^2}{\partial q_j \partial q_k} &= \frac{1}{h_j^2} \frac{\partial^2}{\partial q_j^2} \\ &= \frac{\partial^2}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} \end{aligned}$$

Summing, conclude

$$\begin{aligned}\nabla^2 &= \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\cot(\theta)}{r^2} \frac{\partial}{\partial \theta} + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2}{\partial \phi^2}\end{aligned}$$

(e) Compute (using the definition of $\mathbf{u}$ in part (c))

$$\begin{aligned}(\mathbf{u} \cdot \nabla) \mathbf{u} &= \left[\underbrace{(\mathbf{e}_{q_i} u_{q_i}) \cdot \left(\mathbf{e}_{q_j} \frac{1}{h_j} \frac{\partial}{\partial q_j} \right)}_{(\text{note } \mathbf{e}_{q_i} \cdot \mathbf{e}_{q_j} = \delta_{ij})} \right] \mathbf{e}_{q_k} u_{q_k} \\ &= \frac{u_{q_i}}{h_i} \frac{\partial}{\partial q_i} (\mathbf{e}_{q_k} u_{q_k}) \\ &= \frac{u_{q_i}}{h_i} \left(\frac{\partial \mathbf{e}_{q_k}}{\partial q_i} u_{q_k} + \mathbf{e}_{q_k} \frac{\partial u_{q_k}}{\partial q_i} \right)\end{aligned}$$

Expanding each term:

$$\begin{aligned}\frac{u_{q_i}}{h_i} \frac{\partial \mathbf{e}_{q_k}}{\partial q_i} u_{q_k} &= \mathbf{e}_\theta \frac{u_\theta}{r} u_r + \mathbf{e}_\phi \sin(\theta) \frac{u_\phi}{r \sin(\theta)} u_r - \mathbf{e}_r \frac{u_\theta^2}{r} + \mathbf{e}_\phi \cos(\theta) \frac{u_\phi}{r \sin(\theta)} u_\theta \\ &\quad + (-\mathbf{e}_r \sin(\theta) - \mathbf{e}_\theta \cos(\theta)) \frac{u_\phi}{r \sin(\theta)} u_\phi \\ &= -\mathbf{e}_r \left(\frac{u_\theta^2 + u_\phi^2}{r} \right) \\ &\quad + \mathbf{e}_\theta \left(\frac{u_r u_\theta - u_\phi^2 \cot(\theta)}{r} \right) \\ &\quad + \mathbf{e}_\phi \left(\frac{u_r u_\phi + u_\theta u_\phi \cot(\theta)}{r} \right) \\ \mathbf{e}_{q_k} \frac{u_{q_i}}{h_i} \frac{\partial u_{q_k}}{\partial q_i} &= \mathbf{e}_r (\mathbf{u} \cdot \nabla) u_r + \mathbf{e}_\theta (\mathbf{u} \cdot \nabla) u_\theta + \mathbf{e}_\phi (\mathbf{u} \cdot \nabla) u_\phi\end{aligned}$$

Summing, conclude

$$\begin{aligned}(\mathbf{u} \cdot \nabla) \mathbf{u} &= \mathbf{e}_r \left[(\mathbf{u} \cdot \nabla) u_r - \frac{u_\theta^2 + u_\phi^2}{r} \right] \\ &\quad + \mathbf{e}_\theta \left[(\mathbf{u} \cdot \nabla) u_\theta + \frac{u_r u_\theta - u_\phi^2 \cot(\theta)}{r} \right] \\ &\quad + \mathbf{e}_\phi \left[(\mathbf{u} \cdot \nabla) u_\phi + \frac{u_r u_\phi + u_\theta u_\phi \cot(\theta)}{r} \right]\end{aligned}$$

where

$$\mathbf{u} \cdot \nabla = u_r \frac{\partial}{\partial r} + \frac{u_\theta}{r} \frac{\partial}{\partial \theta} + \frac{u_\phi}{r \sin(\theta)} \frac{\partial}{\partial \phi}$$

Question: 3.3

In a steady two-dimensional flow, Cartesian-component particle trajectories are given by: $x(t) = r_0 \cos(\gamma(t - t_0) + \theta_0)$ and $y(t) = r_0 \sin(\gamma(t - t_0) + \theta_0)$ where $r_0 = \sqrt{x_0^2 + y_0^2}$ and $\theta_0 = \tan^{-1}(y_0/x_0)$.

- (a) From these trajectories determine the Lagrangian particle velocity components $u(t) = dx/dt$ and $v(t) = dy/dt$, and convert these to Eulerian velocity components $u(x, y)$ and $v(x, y)$.
- (b) Compute Cartesian particle acceleration components, $a_x = d^2x/dt^2$ and $a_y = d^2y/dt^2$, and show that they are equal to D/Dt of the Eulerian velocity components $u(x, y)$ and $v(x, y)$.

(a) Differentiate to find

$$\begin{aligned} u(t) &= \frac{dx}{dt} = -\gamma r_0 \sin(\gamma(t - t_0) + \theta_0) \\ &= -\gamma y \\ &= u(x, y) \\ v(t) &= \frac{dy}{dt} = \gamma r_0 \cos(\gamma(t - t_0) + \theta_0) \\ &= \gamma x \\ &= v(x, y) \end{aligned}$$

(b) Differentiate once more to get

$$\begin{aligned} a_x &= \frac{d^2x}{dt^2} = -\gamma^2 r_0 \cos(\gamma(t - t_0) + \theta_0) \\ &= -\gamma^2 x \\ a_y &= \frac{d^2y}{dt^2} = -\gamma^2 r_0 \sin(\gamma(t - t_0) + \theta_0) \\ &= -\gamma^2 y \end{aligned}$$

Define $\mathbf{u} = (u, v)$. The material derivatives are

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u}$$

where each component is

$$\begin{aligned}
 \frac{Du}{Dt} &= \frac{\partial u}{\partial t} + (\mathbf{u} \cdot \nabla) u \\
 &= \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \\
 &= 0 - \gamma y \cdot 0 + \gamma x \cdot (-\gamma) \\
 &= -\gamma^2 x = a_x \\
 \frac{Dv}{Dt} &= \frac{\partial v}{\partial t} + (\mathbf{u} \cdot \nabla) v \\
 &= \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \\
 &= 0 - \gamma y \cdot \gamma + \gamma x \cdot 0 \\
 &= -\gamma^2 y = a_y
 \end{aligned}$$

Since the flow is steady, $\partial \mathbf{u} / \partial t = \mathbf{0}$.

Question: 3.6

For the two-dimensional steady flow having velocity components $u = Sy$ and $v = Sx$, determine the following when S is a positive real constant having units of 1/time.

- equations for the streamlines with a sketch of the flow pattern
- the components of the strain-rate tensor
- the components of the rotation tensor
- the coordinate rotation that diagonalizes the strain-rate tensor, and the principal strain rates
- How is this flow field related to that in Example 3.5.

- Using the tangency requirements:

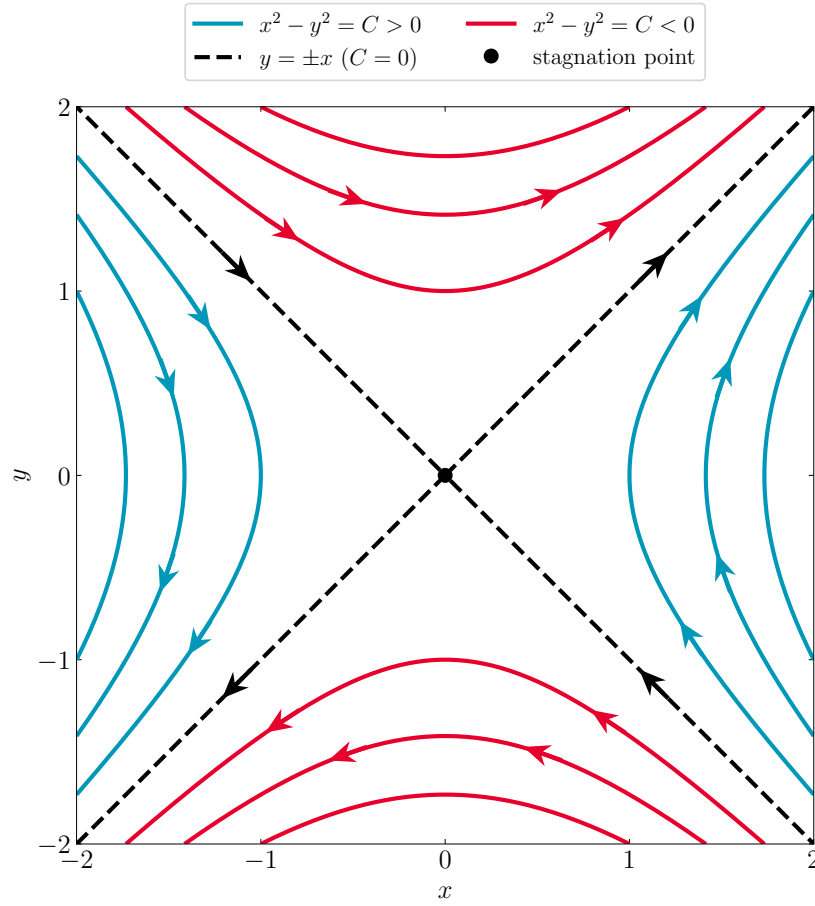
$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}$$

Derive

$$\frac{dy}{dx} = \frac{v}{u} = \frac{Sx}{Sy} = \frac{x}{y}$$

Separating variables and integrating, we find

$$y^2 = x^2 + C$$



(b) Define $\mathbf{u} = (Sy, Sx)$. From the definition of the (symmetric) strain rate tensor

$$S_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

from which we compute

$$S_{ij} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} 0 & S \\ S & 0 \end{bmatrix}$$

(c) From the definition of the (antisymmetric) rotation tensor

$$R_{ij} = \frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}$$

from which we compute

$$R_{ij} = \begin{bmatrix} 0 & \frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

- (d) From the strain rate tensor, compute the determinant as (and set equal to zero)

$$\det|S_{ij} - \lambda\delta_{ij}| = \det\begin{bmatrix} -\lambda & S \\ S & -\lambda \end{bmatrix} = \lambda^2 - S^2 = 0$$

which has solutions $\lambda^{(1)} = S$ and $\lambda^{(2)} = -S$. The corresponding normalized eigenvector solutions are

$$\mathbf{b}^{(1)} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \quad \mathbf{b}^{(2)} = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

Using the eigenvectors, assemble the transformation matrix

$$\mathbf{C} = [\mathbf{b}^{(1)} \quad \mathbf{b}^{(2)}] = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

which corresponds to a 45-degree rotation of the axes. Lastly, derive the transformed strain rate tensor $\mathbf{S}$ in the rotated frame as

$$\mathbf{C}^T \mathbf{S} \mathbf{C} = \mathbf{S}' = \begin{bmatrix} S & 0 \\ 0 & -S \end{bmatrix}$$

- (e) Akin to Example 3.5, the flow lines are hyperbolae, albeit asymptotically bounded by the diagonals rather than by the axes. When oriented around the principal axes and $S > 0$, fluid elements expand in the x' -direction and contract in the y' -direction.

Question: 3.7

At the instant shown in Figure 3.2b, the (u, v) -velocity field in Cartesian coordinates is $u = A(y^2 - x^2)/(x^2 + y^2)^2$, and $v = -2Axy/(x^2 + y^2)^2$ where A is a positive constant. Determine the equations for the streamlines by rearranging the first equality in (3.7) to read $udy - vdx = 0 = (\partial\psi/\partial y) dy + (\partial\psi/\partial x) dx$ and then looking for a solution in the form $\psi(x, y) = \text{const.}$

Taking the tangency requirement $dx/u = dy/v$, multiply by uv and bring the terms to one side to derive

$$udy - vdx = 0$$

By the method of exact equations, we can identify a candidate for ψ if and only if $\partial M/\partial y = \partial N/\partial x$ for $M = -v$ and $N = u$. Compute

$$\begin{aligned} \frac{\partial}{\partial y} \left[\frac{2Axy}{(x^2 + y^2)^2} \right] &= \frac{2Ax}{(x^2 + y^2)^2} - \frac{8Axy^2}{(x^2 + y^2)^3} \\ &= \frac{2A(x^3 - 3xy^2)}{(x^2 + y^2)^3} \\ \frac{\partial}{\partial x} \left[\frac{A(y^2 - x^2)}{(x^2 + y^2)^2} \right] &= -\frac{2Ax}{(x^2 + y^2)^2} - \frac{4Ax(y^2 - x^2)}{(x^2 + y^2)^3} \\ &= \frac{2A(x^3 - 3xy^2)}{(x^2 + y^2)^3} \end{aligned}$$

which matches, ascertaining the existence of ψ . Integrate $-v$ with respect to x and find (by letting $w = x^2 + y^2$ and $dw = 2x dx$)

$$\begin{aligned}\psi(x, y) &= \int \frac{2Axy}{(x^2 + y^2)^2} dx \\ &= Ay \int \frac{dw}{w^2} \\ &= -\frac{Ay}{x^2 + y^2} + g(y)\end{aligned}$$

Differentiate with respect to y to derive

$$\begin{aligned}\frac{\partial \psi}{\partial y} &= -\frac{A}{x^2 + y^2} + \frac{2Ay^2}{(x^2 + y^2)^2} + g'(y) \\ &= \frac{A(y^2 - x^2)}{(x^2 + y^2)^2} + g'(y) \\ &= \frac{A(y^2 - x^2)}{(x^2 + y^2)^2} = u\end{aligned}$$

forcing $g'(y) = 0$ and thus $g(y) = C$ for any constant C . Fix $\psi(x, y) = \psi_0$. Then

$$\psi_0 = C - \frac{Ay}{x^2 + y^2}$$

Multiplying through by $x^2 + y^2$ and rearranging, derive

$$x^2 + \left(y + \frac{\beta}{2}\right)^2 = \left(\frac{\beta}{2}\right)^2, \quad x \neq 0$$

where $\beta = A/(\psi_0 - C)$. The streamlines are then circles with center $(0, -\beta/2)$ and radius $|\beta|/2$. One final step: we have an edge case where $\psi_0 = C$, rendering β undefined. Then we find

$$-\frac{Ay}{x^2 + y^2} = 0 \implies y = 0, \quad x \neq 0$$

Note that due to the $x^2 + y^2$ term in the denominator, the streamlines and the velocities are undefined at the origin $(0, 0)$.

Question: 3.10

Compute and compare the streamline, path line, and streak line that pass through $(1, 1, 0)$ at $t = 0$ for the following Cartesian velocity field $\mathbf{u} = (x, -yt, 0)$.

Streamline. By equation (3.7), derive

$$\frac{dy}{dx} = \frac{v}{u} = -\frac{yt}{x}$$

which, after separating variables and integrating, resolves to

$$y = Cx^{-t}$$

Since $w = 0$, there is no variation in z , leaving only one equation in (x, y) . At $t = 0$, we have $y = C$, requiring $C = 1$. The streamline equation is $y = 1, z = 0$ for $x > 0$; recall from our derivation of dy/dx that x is in the denominator, and the streamline cannot be defined at $x = 0$. Physically, at $t = 0$ and $x = 0$, the velocity is $\mathbf{u} = (0, 0, 0)$, for which no streamlines can exist.

Path-line. Define $\mathbf{r} = [x(t), y(t)]$ and set

$$\frac{dx}{dt} = u = x, \quad \frac{dy}{dt} = v = -yt, \quad \frac{dz}{dt} = w = 0$$

Separate variables and integrate to show

$$x(t) = x_0 e^t, \quad y(t) = y_0 e^{-t^2/2}, \quad z(t) = z_0$$

By the path-line trajectory intercept requirement, $x(0) = x_0 = 1, y(0) = y_0 = 1$, and $z(0) = z_0 = 0$. The path-line component equations are

$$\begin{aligned} x(t) &= e^t \\ y(t) &= e^{-t^2/2} \\ z(t) &= 0 \end{aligned}$$

To do away with the t term, write $t = \ln(x)$ and substitute into $y(t)$:

$$y(x) = e^{-[\ln(x)]^2/2} = x^{-\ln(x)/2}, \quad z = 0, \quad x > 0$$

where once again we must have $x > 0$ as $x(t) = e^t$. Note that at $t = 0$ and $(1, 1, 0)$, $(dx/dt, dy/dt, dz/dt) = (1, 0, 0)$. The velocity is a horizontal vector at that point. As it turns out, the derivative of our path-line at that point is

$$\left. \frac{dy}{dx} \right|_{x=1} = -e^{-[\ln(x)]^2/2} \left[\frac{\ln(x)}{x} \right] = 0$$

which agrees with the direction of our velocity vector. Since the streamline must be tangent to the flow velocity, our result also comports with our streamline equation $y = 1$.

Streak-line. Consider time t_0 which may transpire at any timestep before or after t . Using our equations for x, y , and z for the path-line derivation, we rederive new constants for the fluid particle at $(1, 1, 0)$ at that earlier time:

$$\begin{aligned} x(t_0) = x_0 e^{t_0} = 1 &\implies x_0 = e^{-t_0} \\ y(t_0) = y_0 e^{-t_0^2/2} = 1 &\implies y_0 = e^{t_0^2/2} \\ z(t_0) = 0 &\implies z_0 = 0 \end{aligned}$$

Then substituting back into our component equations, find

$$\begin{aligned} x(t) &= e^{t-t_0} \\ y(t) &= e^{(-t^2+t_0^2)/2} \\ z(t) &= 0 \end{aligned}$$

This time, eliminate t_0 by writing $t_0 = t - \ln(x)$ and substitute

$$y(x) = e^{-t \ln(x) + [\ln(x)]^2/2} = x^{-t + \ln(x)/2}$$

And finally, evaluate at $t = 0$ to conclude

$$y(x) = x^{\ln(x)/2}, \quad z = 0, \quad x > 0$$

We summarize the results in the table below.

Equation	$y(x) (z = 0)$	x -domain
Streamline	1	$x > 0$
Path-line	$x^{-\ln(x)/2}$	$x > 0$
Streak-line	$x^{\ln(x)/2}$	$x > 0$

12 - Turbulence

Question: 12.1

Determine general relationships for the second, third, and fourth central moments (variance = σ^2 , skewness = S , and kurtosis = K) of the random variable u in terms of its first four ordinary moments: $\bar{u}$, $\bar{u}^2$, $\bar{u}^3$, and $\bar{u}^4$.

By definition, the m -th central moment is

$$\overline{(u - \bar{u})^m} = \frac{1}{N} \sum_{n=1}^N \left(u(\mathbf{x}, t : n) - \bar{u}(\mathbf{x}, t) \right)^m$$

Following the rules of operations involving ensemble averaging, we can derive (more succinctly with the binomial expansion¹):

$$\begin{aligned} \sigma^2 &= \overline{(u - \bar{u})^2} \\ &= \overline{u^2 - 2u\bar{u} + \bar{u}^2} \\ &= \bar{u}^2 - 2\bar{u}^2 + \bar{u}^2 \\ &= \boxed{\bar{u}^2 - \bar{u}^2} \\ S &= \overline{(u - \bar{u})^3} \\ &= \overline{u^3 - 3u^2\bar{u} + 3u\bar{u}^2 - \bar{u}^3} \\ &= \bar{u}^3 - 3\bar{u}^2\bar{u} + 3\bar{u}^3 - \bar{u}^3 \\ &= \bar{u}^3 - 3\bar{u}(\bar{u}^2 - \bar{u}^2) - \bar{u}^3 \\ &= \boxed{\bar{u}^3 - 3\bar{u}\sigma^2 - \bar{u}^3} \\ K &= \overline{(u - \bar{u})^4} \\ &= \overline{(u^2 - 2u\bar{u} + \bar{u}^2)(u^2 - 2u\bar{u} + \bar{u}^2)} \\ &= \overline{u^4 - 4u^3\bar{u} + 6u^2\bar{u}^2 - 4u\bar{u}^3 + \bar{u}^4} \\ &= \boxed{\bar{u}^4 - 4\bar{u}^3\bar{u} + 6\bar{u}^2\bar{u}^2 - 3\bar{u}^4} \end{aligned}$$

Question: 12.2

Calculate the mean, mean square, variance, and *rms* value (or standard deviation) of the periodic time series $u(t) = \bar{U} + U_0 \cos(\omega t)$, where $\bar{U}$, U_0 , and ω are positive real constants.

In general, the time average of $u^m(\mathbf{x}, t)$ is given by:

$$\overline{u^m(\mathbf{x})} = \frac{1}{\Delta t} \int_{t-\Delta t/2}^{t+\Delta t/2} u^m(\mathbf{x}, t) dt$$

¹For numbers a, b : $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$

And so we calculate:

Mean:

$$\begin{aligned}
 \bar{u} &= \frac{1}{\Delta t} \int_{t-\Delta t/2}^{t+\Delta t/2} [\bar{U} + U_0 \cos(\omega t)] dt \\
 &= \frac{1}{\Delta t} \left\{ \bar{U} \Delta t + \frac{U_0}{\omega} \left[\sin \left(\omega \left(t + \frac{\Delta t}{2} \right) \right) - \sin \left(\omega \left(t - \frac{\Delta t}{2} \right) \right) \right] \right\} \\
 &= \frac{1}{\Delta t} \left[\bar{U} \Delta t + \frac{2U_0}{\omega} \cos(\omega t) \sin \left(\frac{\omega \Delta t}{2} \right) \right] \\
 &= \boxed{\bar{U} + \left[\frac{\sin(\omega \Delta t/2)}{\omega \Delta t/2} \right] U_0 \cos(\omega t)}
 \end{aligned}$$

Mean Square:

$$\begin{aligned}
 \overline{u^2} &= \frac{1}{\Delta t} \int_{t-\Delta t/2}^{t+\Delta t/2} [\bar{U} + U_0 \cos(\omega t)]^2 dt \\
 &= \frac{1}{\Delta t} \int_{t-\Delta t/2}^{t+\Delta t/2} [\bar{U}^2 + 2\bar{U}U_0 \cos(\omega t) + U_0^2 \cos^2(\omega t)] dt \\
 &= \frac{1}{\Delta t} \left[\bar{U}^2 \Delta t + \frac{2\bar{U}U_0}{\omega} \left(\sin \left(\omega \left(t + \frac{\Delta t}{2} \right) \right) - \sin \left(\omega \left(t - \frac{\Delta t}{2} \right) \right) \right) \right. \\
 &\quad \left. + \frac{U_0^2}{2} \Delta t + \frac{U_0^2}{4\omega} \left(\sin \left(2\omega \left(t + \frac{\Delta t}{2} \right) \right) - \sin \left(2\omega \left(t - \frac{\Delta t}{2} \right) \right) \right) \right] \\
 &= \bar{U}^2 + \frac{U_0^2}{2} + \frac{2\bar{U}U_0}{\omega \Delta t/2} \cos(\omega t) \sin(\omega \Delta t/2) + \frac{U_0^2/2}{\omega \Delta t} \cos(2\omega t) \sin(\omega \Delta t) \\
 &= \boxed{\bar{U}^2 + \frac{U_0^2}{2} + \left[\frac{\sin(\omega \Delta t/2)}{\omega \Delta t/2} \right] 2\bar{U}U_0 \cos(\omega t) + \left[\frac{\sin(\omega \Delta t)}{\omega \Delta t} \right] \frac{U_0^2}{2} \cos(2\omega t)}
 \end{aligned}$$

Squared Mean:

$$\begin{aligned}
 \bar{u}^2 &= \left[\bar{U} + \left[\frac{\sin(\omega \Delta t/2)}{\omega \Delta t/2} \right] U_0 \cos(\omega t) \right]^2 \\
 &= \boxed{\bar{U}^2 + \left[\frac{\sin(\omega \Delta t/2)}{\omega \Delta t/2} \right] 2\bar{U}U_0 \cos(\omega t) + \left[\frac{\sin(\omega \Delta t/2)}{\omega \Delta t/2} \right]^2 U_0^2 \cos^2(\omega t)}
 \end{aligned}$$

Variance:

$$\begin{aligned}
\sigma^2 &= \overline{u^2} - \bar{u}^2 \\
&= \frac{U_0^2}{2} + \left[\frac{\sin(\omega\Delta t)}{\omega\Delta t} \right] \frac{U_0^2}{2} \cos(2\omega t) - \left[\frac{\sin(\omega\Delta t/2)}{\omega\Delta t/2} \right]^2 U_0^2 \cos^2(\omega t) \\
&= \frac{U_0^2}{2} + \left[\frac{\sin(\omega\Delta t)}{\omega\Delta t} \right] \frac{U_0^2}{2} (2\cos^2(\omega t) - 1) - \left[\frac{\sin(\omega\Delta t/2)}{\omega\Delta t/2} \right]^2 U_0^2 \cos^2(\omega t) \\
&= U_0^2 \left[\cos^2(\omega t) \left[\frac{\sin(\omega\Delta t)}{\omega\Delta t} - \left(\frac{\sin(\omega\Delta t/2)}{\omega\Delta t/2} \right)^2 \right] + \frac{1}{2} - \frac{1}{2} \left(\frac{\sin(\omega\Delta t)}{\omega\Delta t} \right) \right]
\end{aligned}$$

Root-Mean-Square:

The authors use the term root-mean-square and standard deviation synonymously here. They are distinct but related:

$$\begin{aligned}
rms &= \sqrt{\overline{U^2} + \frac{U_0^2}{2} + \left[\frac{\sin(\omega\Delta t/2)}{\omega\Delta t/2} \right] 2\bar{U}U_0 \cos(\omega t) + \left[\frac{\sin(\omega\Delta t)}{\omega\Delta t} \right] \frac{U_0^2}{2} \cos(2\omega t)} \\
\sigma &= U_0 \left[\cos^2(\omega t) \left[\frac{\sin(\omega\Delta t)}{\omega\Delta t} - \left(\frac{\sin(\omega\Delta t/2)}{\omega\Delta t/2} \right)^2 \right] + \frac{1}{2} - \frac{1}{2} \left(\frac{\sin(\omega\Delta t)}{\omega\Delta t} \right) \right]^{1/2}
\end{aligned}$$

Observe that as $\Delta t \rightarrow 0$, the bracketed term in the mean goes to unity, recovering the original function – as expected. The intuition here is that decreasing our time interval does not alter the statistics of $u(t)$, indicating that it is stationary in time. Moreover, a smaller time interval forces the variance to vanish. We expect this too, as $u(t)$ will not vary much at all as the neighborhood of points $t \in (t - \Delta t/2, t + \Delta t/2)$ continues to shrink.

Question: 12.3

Show that the autocorrelation function $\overline{u(t)u(t+\tau)}$ of a periodic series $u = U \cos(\omega t)$ is itself periodic.

The correlation function is

$$\overline{u(t)u(t+\tau)} = \frac{1}{\Delta t} \int_{t-\Delta t/2}^{t+\Delta t/2} U^2 \cos(\omega t) \cos(\omega(t+\tau)) dt$$

which evaluates to

$$\begin{aligned}
 \overline{u(t)u(t+\tau)} &= \frac{1}{2\Delta t} \int_{t-\Delta t/2}^{t+\Delta t/2} U^2 [\cos(\omega\tau) + \cos(2\omega t + \omega\tau)] dt \\
 &= \frac{U^2}{2\Delta t} \left[\Delta t \cos(\omega\tau) + \frac{1}{2\omega} \left(\sin \left(2\omega \left(t + \frac{\Delta t}{2} \right) + \omega\tau \right) \right. \right. \\
 &\quad \left. \left. - \sin \left(2\omega \left(t - \frac{\Delta t}{2} \right) + \omega\tau \right) \right) \right] \\
 &= \frac{U^2}{2\Delta t} \left[\Delta t \cos(\omega\tau) + \frac{1}{2\omega} (2 \cos(2\omega t + \omega\tau) \sin(2\omega\Delta t)) \right] \\
 &= U^2 \left[\frac{1}{2} \cos(\omega\tau) + \cos(2\omega t + \omega\tau) \left[\frac{\sin(2\omega\Delta t)}{2\omega\Delta t} \right] \right]
 \end{aligned}$$

and is periodic – the only term with t is the second cosine term.

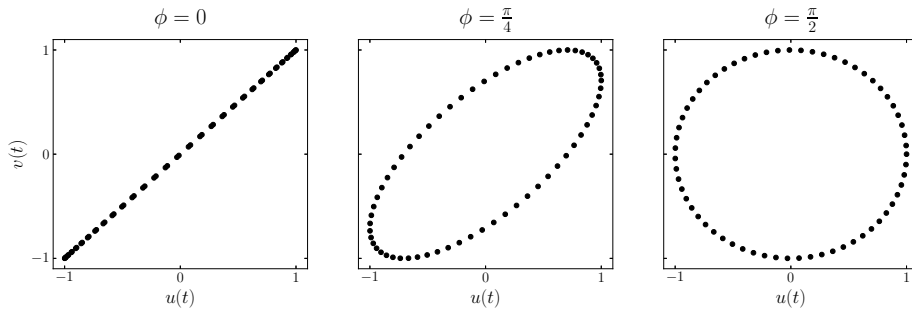
Question: 12.4

Calculate the zero-lag cross-correlation $\overline{u(t)v(t)}$ between two periodic series $u(t) = \cos(\omega t)$ and $v(t) = \cos(\omega t + \phi)$ by performing a time average over one period $= 2\pi/\omega$. For values of $\phi = 0, \pi/4$, and $\pi/2$, plot the scatter diagrams of u vs v at different times, as in Figure 12.8. Note that the plot is a straight line if $\phi = 0$, an ellipse if $\phi = \pi/4$, and a circle if $\phi = \pi/2$; the straight line, as well as the axes of the ellipse, are inclined at 45° to the uv -axes. Argue that the straight line signifies a perfect correlation, the ellipse a partial correlation, and the circle a zero correlation.

The zero-lag cross-correlation is given by

$$\begin{aligned}
 \overline{u(t)v(t)} &= \frac{\omega}{2\pi} \int_{t-\pi/\omega}^{t+\pi/\omega} \cos(\omega t) \cos(\omega t + \phi) dt \\
 &= \frac{\omega}{2\pi} \int_{t-\pi/\omega}^{t+\pi/\omega} \frac{1}{2} [\cos(\phi) + \cos(2\omega t + \phi)] dt \\
 &= \frac{\omega}{4\pi} \left[\frac{2\pi}{\omega} \cos(\phi) \right] \\
 &= \boxed{\frac{1}{2} \cos(\phi)}
 \end{aligned}$$

When $\phi = 0, \pi/4, \pi/2$, the cross-correlation is $1/2, 1/2\sqrt{2}, 0$, respectively. Parametrically plotting $u(t)$ and $v(t)$, we see that the values of ϕ correspond to a straight line, ellipse, and circle, respectively:

**Question: 12.5**

If $u(t)$ is a stationary random signal, show that $u(t)$ and $du(t)/dt$ are uncorrelated.

The cross-correlation is given by:

$$\begin{aligned} \overline{u(t) u'(t)} &= \frac{1}{\Delta t} \int_{t-\Delta t/2}^{t+\Delta t/2} u(t) \frac{du}{dt} dt \\ &= \frac{1}{\Delta t} \int_{t-\Delta t/2}^{t+\Delta t/2} u(t) du \\ &= \frac{1}{2\Delta t} \left(u\left(t + \frac{\Delta t}{2}\right)^2 - u\left(t - \frac{\Delta t}{2}\right)^2 \right) \end{aligned}$$

Truthfully, I'm quite confused as to what du/dt is under the premise that $u(t)$ is stochastic. My belief is that the time derivative would not exist normally for a stochastic process?

Question: 12.6

Let $R(\tau)$ and $S(\omega)$ be a Fourier transform pair. Show that $S(\omega)$ is real and symmetric if $R(\tau)$ is real and symmetric.

Suppose that $R(\tau)$ is real and symmetric. Then $R(\tau) = R(-\tau)$. Then we define $S(\omega)$ as

$$\begin{aligned} S(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} R(\tau) \exp(-i\omega\tau) d\tau \\ &= \frac{1}{2\pi} \left[\int_{-\infty}^{+\infty} R(\tau) \cos(\omega\tau) d\tau - i \int_{-\infty}^{+\infty} R(\tau) \sin(\omega\tau) d\tau \right] \end{aligned}$$

By realness of $R(\tau)$, the first term must be real. Moreover, by symmetry of $R(\tau)$, its integrand is even, and so its integral is also symmetric. Once more by symmetry, the integrand of the second function is odd, and its integral thus vanishes. All that remains is

$$S(\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} R(\tau) \cos(\omega\tau) d\tau$$

which demonstrates that $S(\omega)$ is real and symmetric.

Question: 12.7

Compute the power spectrum, integral time scale, and Taylor time scale when $R_{11}(\tau) = \overline{u_1^2} \exp(-\alpha\tau^2) \cos(\omega_0\tau)$, assuming that α and ω_0 are real positive constants.

Power Spectrum:

$$\begin{aligned} S(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left[\overline{u_1^2} \exp(-\alpha\tau^2) \cos(\omega_0\tau) \right] \exp(-i\omega_0\tau) d\tau \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \overline{u_1^2} \exp(-\alpha\tau^2 - i\omega_0\tau) \cos(\omega_0\tau) d\tau \end{aligned}$$

Now, invoking the identity

$$\cos(\omega_0\tau) = \frac{1}{2}(\exp(i\omega_0\tau) + \exp(-i\omega_0\tau))$$

we can write

$$\begin{aligned} &= \frac{1}{4\pi} \int_{-\infty}^{+\infty} \overline{u_1^2} \exp(-\alpha\tau^2 - i\omega_0\tau) (\exp(i\omega_0\tau) + \exp(-i\omega_0\tau)) d\tau \\ &= \frac{\overline{u_1^2}}{4\pi} \left[\int_{-\infty}^{+\infty} \exp(-\alpha\tau^2) d\tau + \int_{-\infty}^{+\infty} \exp(-\alpha\tau^2 - 2i\omega_0\tau) d\tau \right] \\ &= \frac{\overline{u_1^2}}{4\pi} \left[\sqrt{\frac{\pi}{\alpha}} + \sqrt{\frac{\pi}{\alpha}} \exp\left(-\frac{4\omega_0^2}{4\alpha}\right) \right] \\ &= \boxed{\frac{\overline{u_1^2}}{4\pi} \sqrt{\frac{\pi}{\alpha}} \left[1 + \exp\left(-\frac{\omega_0^2}{\alpha}\right) \right]} \end{aligned}$$

Observe that, consistent with the result of question 12.6, as $R_{11}(\tau)$ is real and symmetric, $S(\omega)$ is real and symmetric.

Integral Time Scale

$$\begin{aligned}
\Lambda_t &= \frac{1}{R_{11}(0)} \int_0^\infty R_{11}(\tau) d\tau \\
&= \frac{1}{R_{11}(0)} \int_0^\infty \overline{u_1^2} \exp(-\alpha\tau^2) \cos(\omega_0\tau) d\tau \\
&= \frac{1}{2R_{11}(0)} \int_{-\infty}^{+\infty} \overline{u_1^2} \exp(-\alpha\tau^2) \cos(\omega_0\tau) d\tau \\
&= \frac{\overline{u_1^2}}{4R_{11}(0)} \left[\int_{-\infty}^{+\infty} \exp(-\alpha\tau^2 + i\omega_0\tau) d\tau + \int_{-\infty}^{+\infty} \exp(-\alpha\tau^2 - i\omega_0\tau) d\tau \right] \\
&= \frac{\overline{u_1^2}}{2R_{11}(0)} \sqrt{\frac{\pi}{\alpha}} \exp\left(-\frac{\omega_0^2}{4\alpha}\right)
\end{aligned}$$

where at the third equality, we rely on the symmetry of the integrand. Now, observe that

$$R_{11}(0) = \overline{u_1^2}$$

and so we can conclude

$$\Lambda_t = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}} \exp\left(-\frac{\omega_0^2}{4\alpha}\right)$$

Taylor Time Scale

Begin with the definition $r_{11}(\tau) = R_{11}(\tau) / R_{11}(0)$. Then

$$r_{11}(\tau) = \exp(-\alpha\tau^2) \cos(\omega_0\tau)$$

Breaking $r_{11}(\tau)$ into its terms and taking the Taylor series:

$$\begin{aligned}
\exp(-\alpha\tau^2) &\approx 1 - \alpha\tau^2 \\
\cos(\omega_0\tau) &\approx 1 - \frac{\omega_0^2\tau^2}{2}
\end{aligned}$$

Keeping the product to two terms, we find

$$r_{11}(\tau) \approx 1 - \left(\frac{\omega_0^2}{2} + \alpha\right) \tau^2$$

which gives us second derivative

$$\frac{d^2 r_{11}}{d\tau^2}(\tau) \approx -(\omega_0^2 + 2\alpha)$$

and lastly, the Taylor time scale

$$\lambda_t = \left(\frac{2}{\omega_0^2 + 2\alpha}\right)^{1/2}$$

Question: 12.8

Two formulae for the energy spectrum $S_e(\omega)$ of the stationary zero-mean signal $u(t)$ are:

$$S_e(\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} R_{11}(\tau) \exp(-i\omega\tau) d\tau$$

and

$$S_e(\omega) = \lim_{T \rightarrow \infty} \frac{1}{2\pi T} \left| \int_{-T/2}^{+T/2} u(t) \exp(-i\omega t) dt \right|^2$$

Prove that these two are identical *without* requiring the existence of the Fourier transform of $u(t)$.